



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation

A Software Engineering Project Submitted

By

Semester: Summer 2024-2025		Section: D	Group Number: 01	
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The project will be evaluated for the following Course Outcomes

CO3 (PO-g-1) <i>Select appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects</i>	Total Marks	
Selection of Software Engineering Models: Process model selection and presents sufficient evidence to support argument for the model selection	[5Marks]	
Role identification and Responsibility Allocation: Well-planned project with proper role identification and responsibility allocation in the project management activities	[5Marks]	
Formation and Submission: Submission, Defense, Completeness, Spelling, grammar, and Organization of the Project report	[5Marks]	
Impact identification: Analysis of the impact of societal, health, safety, legal, and cultural issues	[5Marks]	
CO4 (PO-k-1) <i>Apply engineering management principles and economic decision making to develop software engineering project management plan.</i>	Total Marks	
Project WBS and Testcases: Relevant WBS (project task list) and testcases for the proposed project are stated properly.	[5Marks]	
Effort Estimation and Scheduling: Project estimation was described using proper effort estimation or schedules based on available project resources	[5Marks]	
Risk Management: Sufficient and appropriate risks are identified, analyzed, and properly categorized or prioritized.	[5Marks]	

PROJECT TITLE

AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation

1. PROJECT PROPOSAL

1.1 Background to the Problem

The American International University-Bangladesh (AIUB) Portal (<https://portal.aiub.edu/>) serves as the primary digital interface for students, faculty, and administrators. However, inefficiencies in course registration, campus navigation, resource access, communication, printing, complaint handling, gymnasium services, file sharing, and class scheduling lead to delays, user frustration, reduced engagement, and administrative bottlenecks. These issues undermine AIUB's vision of a technologically advanced, student-centric campus, necessitating a comprehensive digital overhaul.

1.2 Solution to the Problem and Process Model Selection

1.2.1 Objective

The enhanced AIUB student portal intends to enhance the entire student experience by integrating academic, administrative, and personal support services on a single platform. The new features include addressed by AIUB Portal Redefined

- CGPA & Demand-Based Course Registration System
- Lost and Found Submission
- Smart Location System (AI + AR Navigation)
- Event and Notice Display
- Gymnasium Registration & Payment
- Anonymous Complaint System
- Course Book Reference & Online Resources
- Global Printing System
- Faculty Information Directory
- AIUB File Sharing System
- Class Routine & Upcoming Registration Course Generator

1.2.2 Analysis of Software Nature and Environment

- **Project Nature:** The "AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation" is a complex, multi-module system integrating AI-driven course registration, AR-based navigation, cloud printing, anonymous complaints, and file sharing. It serves a diverse user base (students, faculty, administrators) in a dynamic university environment, requiring high interactivity and adaptability.

- **Environment Characteristics:**
 - **Dynamic Requirements:** Features like CGPA-based registration, AR navigation, and anonymous complaints may evolve based on user feedback and institutional needs.
 - **Technological Complexity:** Combines React.js, Node.js, MySQL, ARKit/ARCore, TensorFlow, AWS, and BLE beacons, necessitating iterative integration and testing.
 - **User-Centric Design:** Requires continuous feedback to ensure usability (e.g., intuitive navigation, accessible interfaces) and inclusivity (e.g., wheelchair-friendly routes).
 - **Scalability Needs:** Must handle peak loads (e.g., registration periods) with >99% uptime, as specified in the SRS non-functional requirements.
 - **Societal and Legal Considerations:** Anonymous complaints and data privacy (HTTPS, encryption) address safety and legal compliance, while accessibility modes support inclusivity.
 - **Budget and Timeline:** Operates within AIUB's academic budget and a 16-month timeline (SRS: 3+6+4+3 months), requiring cost-effective, modular development.

1.2.3 Selected Process Model: SCRUM

- **Rationale:** The existing SCRUM model (as per "SE - Project Process Model [OBE].docx") is well-suited due to its iterative, user-focused approach, which aligns with the project's dynamic requirements, complex technology stack, and need for continuous stakeholder engagement. However, the documentation can be enhanced with more precise arguments, evidence, and impact analysis to meet the evaluation criteria comprehensively.

1.2.4 Arguments and Reasons for SCRUM Selection

- **Adaptability:** SCRUM's sprints enable rapid adjustments to evolving needs like course registration and AR navigation, unlike Waterfall's rigid phases.
- **Technology Integration:** Iterative sprints support parallel development and testing of complex modules (e.g., BLE beacons, payment gateways), unlike V-Model's sequential approach.
- **User-Centric:** Sprint reviews ensure usability and accessibility (e.g., <2-second response, wheelchair-friendly routes), unlike Spiral's risk focus.
- **Incremental Delivery:** Early module delivery (e.g., registration) aligns with SRS phased deployment, unlike Waterfall's delayed release.
- **Scalability:** Incremental releases test scalability for peak loads (>99% uptime), per SRS requirements.
- **Risk Mitigation:** Daily standups catch issues early (e.g., payment bugs), unlike Waterfall's late discovery.

- **AIUB Alignment:** SCRUM delivers a scalable, user-centric platform, meeting SRS performance goals and AIUB's digital transformation vision.

1.2.5 Evidence Supporting SCRUM Selection for AIUB Portal Redefined

The SCRUM model is the most suitable for the "AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation" project, as it aligns with the dynamic, complex, and user-centric nature of the system. Below is a detailed comparison with other process models, supported by specific evidence from the project proposal and SRS, focusing on features like CGPA & Demand-Based Course Registration System, Smart Location System (AI + AR Navigation), Global Printing System, Anonymous Complaint System, and others.

1. Compared to Waterfall:

- **Evidence:** The project proposal highlights the need for a flexible system to address evolving needs, such as CGPA-based course registration and AR navigation, which may change due to user feedback or new university policies (e.g., accessibility for disabled users). Waterfall's linear structure locks phases (e.g., planning, design), making changes costly and difficult. SCRUM's 2–3-week sprints, as outlined in the SRS implementation phases, enable adaptive planning, allowing adjustments to features like course registration waitlists or AR navigation prompts based on real-time feedback, ensuring alignment with AIUB's student-centric goals.

2. Compared to V-Model:

- **Evidence:** The V-Model's heavy focus on verification and validation per phase is time-consuming and suited for fixed requirements. The SRS specifies features like real-time schedule conflict detection in course registration and encrypted anonymous complaints, which require iterative user validation to ensure usability and security. SCRUM's sprint reviews, planned in the SRS pilot testing phase, facilitate continuous feedback from students and faculty, allowing features like event notices and gym payments to evolve with semester-based needs, unlike the V-Model's rigid structure.

3. Compared to Prototyping Model:

- **Evidence:** The Prototyping Model excels at visualizing ideas but lacks structure for large systems. The project proposal outlines 11 diverse features, including a searchable lost and found database and faculty directory, requiring scalability and maintainability. SCRUM's structured sprints and roles (e.g., Product Owner for backlog management), as per the SRS iterative development phase, ensure disciplined delivery of complex modules like AI-driven course demand analysis and AR overlays with BLE beacons, unlike the unstructured Prototyping Model.

4. Compared to Incremental Development:

- **Evidence:** Incremental Development delivers system parts over time but lacks SCRUM's collaborative framework. The SRS prioritizes features like course

registration and global printing, requiring stakeholder feedback from students and admins. SCRUM's organized sprints, daily standups, and reviews, as detailed in the SRS pilot testing, enhance cooperation and refine features like the waitlist mechanism and cloud printing notifications, ensuring agility and progress, unlike Incremental Development's limited focus on collaboration.

5. Compared to Component-Based Development:

- **Evidence:** Component-Based Development emphasizes reusable components, ideal for modularity but less adaptive to changing needs. The project proposal highlights dynamic features like anonymous complaints and course resources, which need ongoing user-driven updates. SCRUM's iterative cycles, as per the SRS pilot testing phase, support continuous refinement of features like wheelchair-friendly navigation routes and secure file sharing links, aligning with AIUB's inclusivity goals, unlike Component-Based Development's static focus.

6. Compared to RAD (Rapid Application Development):

- **Evidence:** RAD focuses on quick prototyping with minimal documentation, suitable for small projects but inadequate for the complex system described in the proposal (e.g., 11 integrated features). The SRS requires scalability (>99% uptime) and security (HTTPS, encryption) for features like global printing and anonymous complaints. SCRUM's disciplined framework, with sprint reviews and backlog prioritization, ensures quality and scalability, as outlined in the SRS deployment phase, unlike RAD's lack of structure for multi-module systems.

7. Compared to RUP (Rational Unified Process):

- **Evidence:** RUP's process-heavy phases (e.g., Inception, Construction) suit large enterprise systems but are slow for academic contexts. The proposal emphasizes rapid adaptation for features like event notices and gymnasium registration. SCRUM's lightweight sprints, aligned with the SRS's 16-month timeline, support quick iterations and stakeholder feedback, ensuring features like course resource access and faculty directory updates meet AIUB's dynamic needs, unlike RUP's complex structure.

8. Compared to Basic Agile:

- **Evidence:** Basic Agile offers flexibility but lacks a defined framework. The SRS specifies structured development with roles (e.g., Product Owner, Scrum Master) and artifacts (e.g., product backlog), aligning with SCRUM's events like sprint planning. This ensures measurable progress for features like AI-driven course suggestions and AR navigation with voice guidance, making SCRUM more actionable than Basic Agile for AIUB's multi-stakeholder environment.

9. Compared to XP (Extreme Programming):

- **Evidence:** XP prioritizes coding practices, ideal for small teams but less effective for stakeholder coordination. The proposal requires collaboration on features like faculty directory and anonymous complaints. SCRUM's roles and sprint reviews,

as per the SRS pilot testing, ensure stakeholder input from students, faculty, and admins, delivering a comprehensive system with features like secure file sharing and print usage reports, unlike XP's technical focus.

10. Alignment with Project Goals and SRS Requirements:

- **Evidence:** The proposal underscores AIUB's digital transformation vision, requiring a scalable, user-centric platform with features like CGPA-based registration, AR navigation, and cloud printing. SCRUM's iterative cycles align with the SRS's phased approach (Research, Development, Pilot Testing, Deployment), enabling early delivery of high-priority modules (e.g., registration, printing) while refining others (e.g., file sharing). The SRS's non-functional requirements (<2-second response, >99% uptime) are supported by SCRUM's continuous testing, ensuring performance and reliability across all features.

1.2.6 Impact Analysis on Societal, Health, Safety, Legal, and Cultural Issues

- **Societal Impact:**
 - **Equitable Education:** CGPA-based registration ensures fair course access, addressing social equity.
 - **Inclusivity:** Accessibility modes in AR navigation support disabled users, promoting equal access.
- **Health Impact:**
 - **Student Wellness:** Online gymnasium registration encourages fitness, enhancing student health.
 - **Mental Health:** Streamlined registration and navigation reduce stress from manual processes.
- **Safety Impact:**
 - **Anonymous Complaints:** Encrypted submissions encourage reporting of harassment, improving campus safety.
 - **Navigation Safety:** AR navigation reduces confusion, minimizing risks in crowded campus areas.
- **Legal Impact:**
 - **Data Privacy:** HTTPS, RBAC, and encryption (SRS non-functional requirements) ensure compliance with data protection laws.
 - **Auditability:** Time-stamped registration and printing logs support legal transparency.
- **Cultural Impact:**
 - **Community Engagement:** Centralized event notices foster participation in cultural and academic activities.
 - **Ethical Sensitivity:** SCRUM's stakeholder feedback ensures alignment with AIUB's cultural values, such as inclusivity and transparency.

1.3 Project Role Identification and Responsibilities

1.3.1 Identified Responsibilities of the Roles and Stakeholders

- **Product Owner:**
 - **Responsibilities:**
 - Gathers requirements from AIUB stakeholders (students, faculty, admins).
 - Prioritizes features (e.g., high-priority registration, printing; medium-priority navigation, complaints).
 - Defines user stories with acceptance criteria (e.g., “Students can register without conflicts”).
 - Ensures alignment with AIUB’s academic and administrative goals.
- **Scrum Master:**
 - **Responsibilities:**
 - Facilitates sprint planning, daily standups, reviews, and retrospectives.
 - Removes blockers (e.g., delays in AWS setup, stakeholder feedback delays).
 - Ensures adherence to SCRUM principles and timely delivery of increments.
 - Coaches team on Agile practices to maintain productivity and quality.
- **Software Developers:**
 - **Responsibilities:**
 - Develop frontend (React.js) for intuitive interfaces (e.g., registration forms, navigation maps).
 - Build backend (Node.js) and database (MySQL/SQL Server) for data management.
 - Integrate APIs for payment gateways (bKash) and file sharing.
- **AI/ML Engineers:**
 - **Responsibilities:**
 - Train AI models (TensorFlow, Scikit-learn) for course demand analysis and AR route optimization.
 - Ensure model accuracy (e.g., >95% for demand predictions, per SRS evaluation metrics).
- **AR Developers:**
 - **Responsibilities:**
 - Implement AR navigation using ARKit, ARCore, and Navigine SDK.
 - Integrate BLE beacons and QR code markers for precise indoor positioning (<2-second response).
- **UI/UX Designers:**
 - **Responsibilities:**
 - Design mobile-friendly, accessible interfaces for registration, navigation, and dashboards.
 - Ensure inclusivity with wheelchair-friendly navigation modes and intuitive layouts.

- **QA/Test Engineers:**
 - **Responsibilities:**
 - Conduct unit, integration, and user acceptance testing for all modules.
 - Validate AR navigation accuracy, registration logic, and payment security.
- **DevOps Engineers:**
 - **Responsibilities:**
 - Manage AWS deployment, CI/CD pipelines, and system monitoring.
 - Ensure scalability (e.g., handling peak registration loads) and >99% uptime.
- **System Administrators:**
 - **Responsibilities:**
 - Maintain security (HTTPS, RBAC, encryption for complaints).
 - Update databases (faculty directory, course resources) and generate print usage reports.
- **Customers/Stakeholders (Students, Faculty, Admins):**
 - **Responsibilities:**
 - Provide feedback during sprint reviews on usability (e.g., navigation ease, complaint system).
 - Validate features in pilot testing (SRS Phase 3) and suggest improvements.
 - Clarify requirements via the Product Owner (e.g., event notice filters, resource access).
- **Management (AIUB Leadership):**
 - **Responsibilities:**
 - Allocate budget, infrastructure, and resources (e.g., AWS servers, BLE beacons).
 - Align project with AIUB's strategic goals (digital transformation, student satisfaction).
 - Support stakeholder engagement and remove organizational barriers.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

2.1 Functional Requirements

2.I. CGPA & Demand-Based Course Registration System

1. The system shall allow students to register for courses based on **batch priority** and **CGPA slots** to ensure fair access to high-demand courses.
2. CGPA-based registration slots shall be defined as:
 - **High CGPA (3.50 – 4.00)**: First slot
 - **Mid CGPA (3.00 – 3.49)**: Second slot
 - **Low CGPA (< 3.00)**: Final slot
3. The system shall open registration **batch-by-batch** with fixed time windows to prevent server overload.
4. Students must submit a **pre-registration form** before midterm, including:
 - Current CGPA
 - Completed credits
 - Courses desired for next semester
 - Credit limit preference
5. The system shall automatically analyze **student demand** and open course sections accordingly.
6. The system shall **detect and prevent scheduling conflicts** in real-time during registration.
7. The system shall implement a **waitlist mechanism**:
 - If a section is full, students can join a waitlist.
 - When a seat is available, the system automatically allocates it based on waitlist priority.
8. The system shall allow students to submit **course priority requests** if they cannot register for required subjects.
9. Advisors shall be able to **approve and adjust registrations online**, removing the need for physical queues.
10. All registration activities shall be **time-to-time** for audit and dispute resolution.

Priority Level: High

Precondition: Students must be logged into the system.

2.II. Lost and Found Submission

1. Users shall be able to submit reports of lost or found items with a detailed description and images.

2. The submission shall be stored in a searchable database with filters (date, category, location).
3. Admin approval is required before the post becomes publicly visible.
4. Users shall be able to update or remove their submissions.

Priority Level: High

Precondition: Valid access credentials.

2.III. Smart Location System (AI + AR Navigation)

1. The system shall provide a **real-time interactive map** of AIUB's campus including **all buildings, floors, and room numbers**.
2. The location system shall integrate:
 - **Indoor Positioning System (IPS)** using Wi-Fi RTT, BLE beacons, or UWB.
 - **AI-based image recognition** to identify building markers and signs via camera.
 - **Augmented Reality (AR)** overlays to show directional arrows in the live camera feed.
3. Users shall be able to search for:
 - Classrooms (e.g., "C-324")
 - Faculty offices
 - Labs, library, gym, event halls
4. The system shall display **step-by-step guidance**:
 - Turn-by-turn navigation inside buildings
 - "Up/Down" indicators for stairs or elevators
5. The AR overlay shall:
 - Show arrows for left/right turns
 - Indicate distance to the next step
 - Provide text prompts ("Go Upstairs", "Turn Left")
6. **Voice guidance** shall be available for navigation.
7. The system should have an **accessibility mode** for wheelchair-friendly routes.
8. The map database shall be **regularly updated** to reflect room relocations or building changes.
9. The system shall work in **offline mode** inside campus after the initial map is caught.

Priority Level: Medium

Precondition: Users must access the map interface; camera and location permissions must be granted.

2.IV. Event and Notice Display

1. The system shall display **categorized notices** and **events** for previous, current, and upcoming dates.
2. Users shall be able to filter notices by **type, date, and audience**.
3. Students can register for **competitions, seminars, workshops** directly from the notice board.
4. Only authorized admins shall publish or modify notices.

Priority Level: Medium

Precondition: Admin publishing access.

2.V. Gymnasium Registration & Payment

1. Students can register for gym memberships online.
2. Payment tracking shall be integrated with the system.
3. Payment confirmation shall require admin verification.
4. Students can view membership history and renewal dates.

Priority Level: Medium

Precondition: Logged-in student account.

2.VI. Anonymous Complaint System

1. Students can submit complaints anonymously without revealing their identity.
2. Complaint data shall be encrypted and only accessible to higher authorities.
3. The system allow status tracking for submitted complaints without exposing identity.

Priority Level: Medium

Precondition: Anonymous submission form enabled.

2.VII. Course Book Reference & Online Resources

1. Authorities shall maintain a section with **reference books** for all courses.
2. Each entry shall include book details and links to **soft copies or online resources** (if legally permissible).
3. Students can search and download available resources.

Priority Level: Medium

Precondition: Authorities upload book details and resources.

2.VIII. Global Printing System

1. The system shall allow students to **upload documents remotely** through the portal from any device.
2. Students can select:
 - Printer location (any AIUB campus building)
 - Paper size & type
 - Number of copies
 - Color or grayscale printing
3. Documents shall be stored **securely in a print queue** until the user authenticates at the printer.
4. Printing shall require **student ID card scanning** or secure PIN entry at the printer.
5. The system shall integrate with a **cloud print management tool** similar to “SafeQ” to:
 - Enable global printing from any location
 - Prevent dependency on specific AIUB PCs
6. The system should notify students when documents are **ready for pickup**.
7. Payment options shall include:
 - Deduction from student account balance
 - Online banking payment
8. Admins shall have access to **print usage reports** for tracking and auditing.

Priority Level: High

Precondition: Students must be logged in; license and system updates must be applied.

2.IX. Faculty Information Directory

1. Searchable profiles of faculty members with department, contact info, office location, and teaching courses.
2. Faculty profiles shall also list research areas and publications.
3. Directory updates shall be managed via the admin panel.

Priority Level: Medium

Precondition: Directory database initialized.

2.X. AIUB File Sharing System

1. Users can send and receive files within the AIUB network through the portal.
2. The system shall support temporary **secure links** for file sharing.
3. Files shall auto-delete after an expiry period to save storage and ensure privacy.

Priority Level: Low

Precondition: Authenticated AIUB network user

2.XI. Class Routine & Upcoming Registration Course Generator

1. The system shall detect schedule conflicts during registration.
2. It should suggest eligible courses for the upcoming semester based on prerequisites.
3. Students can generate a **personalized routine** with preferred time slots.

Priority Level: Medium

Precondition: Logged-in student account.

2.2 Non-Functional Requirements

- **Performance:** Respond to user actions within 2 seconds.
- **Security:** Use HTTPS, role-based access control, and encryption for anonymous complaints.
- **Usability:** Mobile-friendly, intuitive, accessible interface supporting disabilities.
- **Reliability:** Ensure high availability with minimal downtime; perform daily backups.
- **Scalability:** Support increased user load during peak times (e.g., registration).

3. PROJECT ESTIMATION AND SCHEDULING

3.1 Effort and Cost Estimation

Project Estimation(COCOMO) :

The AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation project has adopted the Organic COCOMO (Constructive Cost Model) mode due to the well-defined project requirements and manageable system complexity.

This mode is deemed most suitable as it facilitates realistic estimations of effort and resources for projects characterized by structured objectives and relatively low innovation risk. The Organic COCOMO model is specifically tailored for projects where the development team is experienced, the project scope is clearly delineated, and the technical challenges are within the team's expertise, aligning with the nature of the AIUB Portal Redefined initiative.

The application of the Organic COCOMO model is also informed by the project's timeline and resource constraints, as evidenced by the test execution dates ranging from May 1 to September 30, 2025.

This short development window necessitates a model that provides reliable effort and schedule estimates, which the Organic COCOMO achieves by leveraging historical data from similar projects.

For **Organic** type,

P : project complexity = 1.05, **T** : SLOC-dependent coefficient = 0.38,

SLOC : source lines of code = 2450 lines (2.45 KSLOC) , **Coefficient**<Effort Factor> = 2.4

Effort = PM = Coefficient<Effort Factor>***(SLOC/1000)^P**

$$= 2.4 * (2450 / 1000)^{1.05}$$

$$= 6.25 \text{ Person-Months}$$

Development time = DM = 2.50*(PM)^T

$$= 2.50 * (6.25)^{0.38}$$

$$= 5.0 \text{ Months}$$

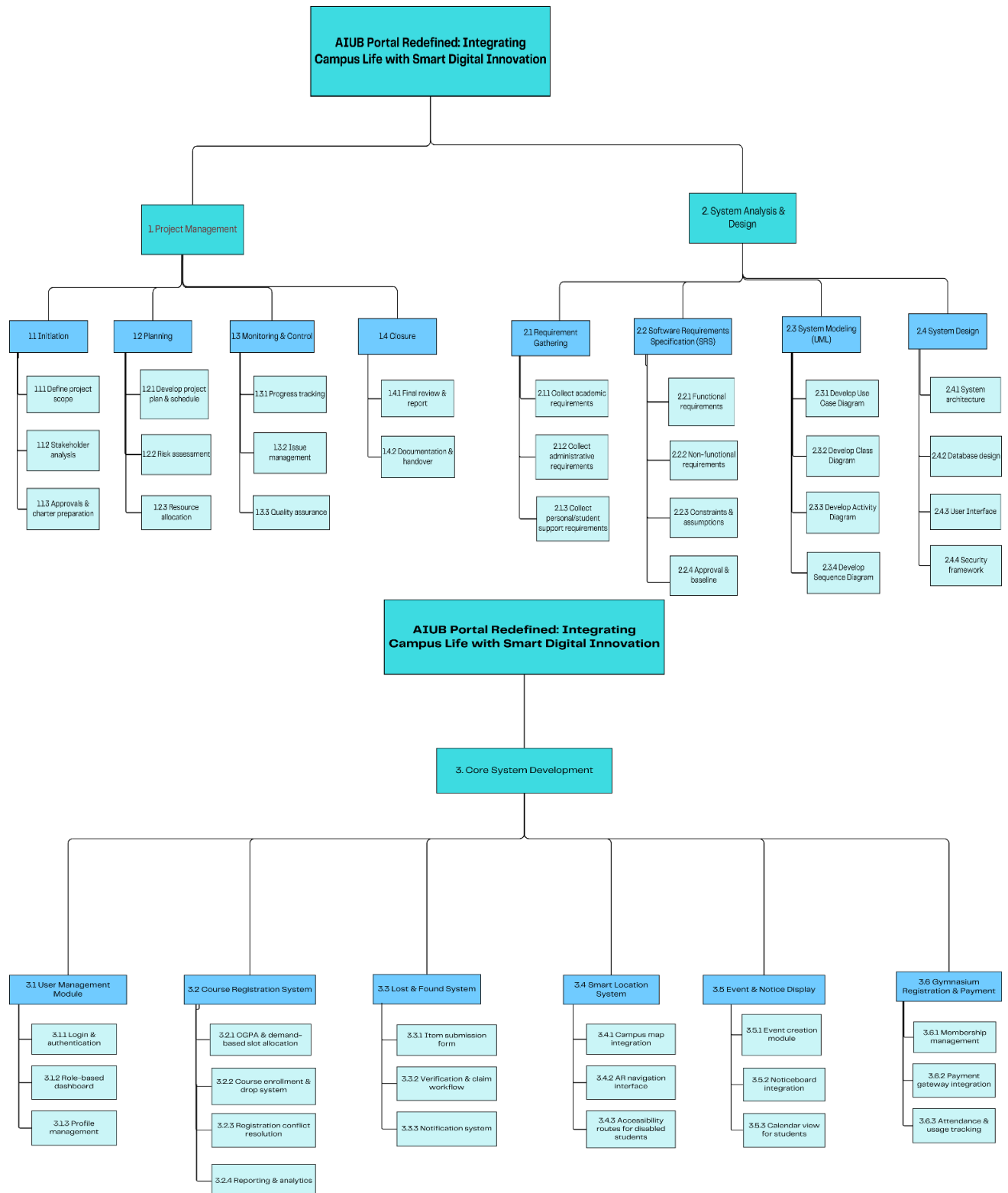
Required number of people = ST = PM/DM

$$= 6.25 / 5.0$$

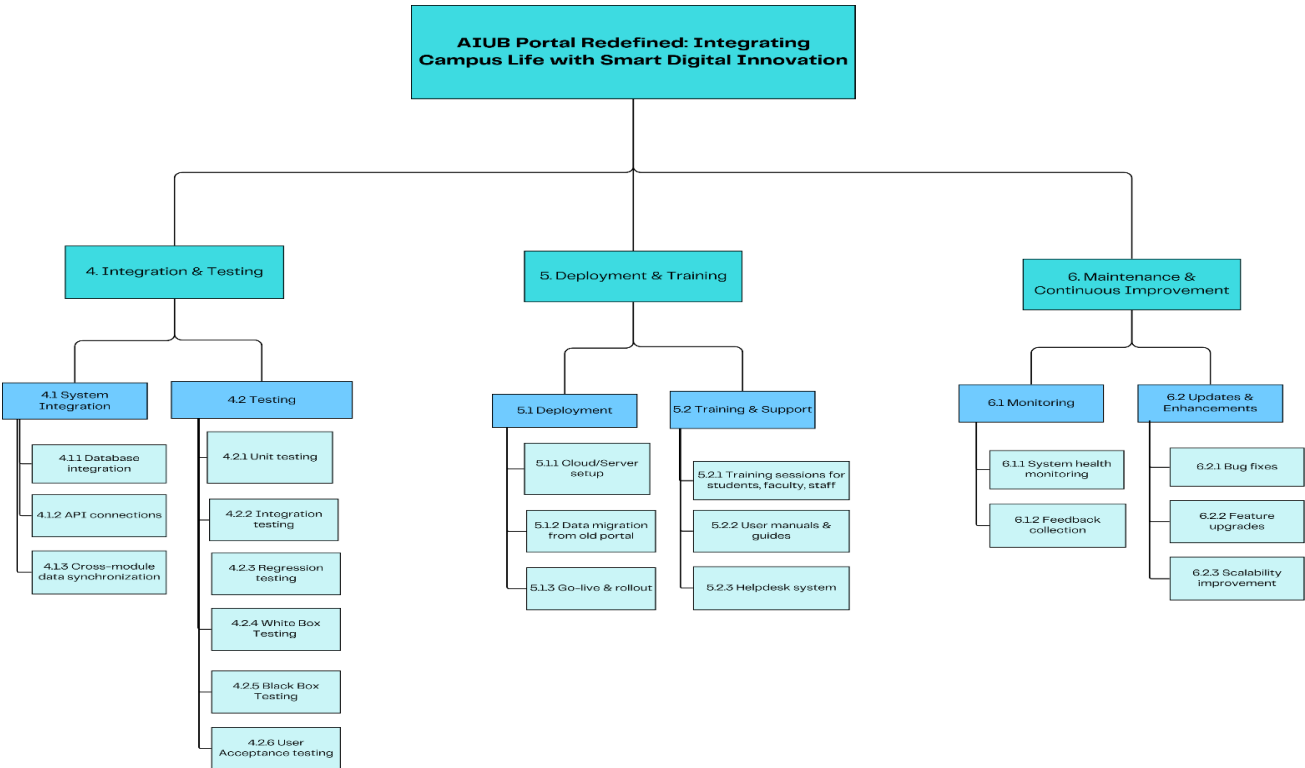
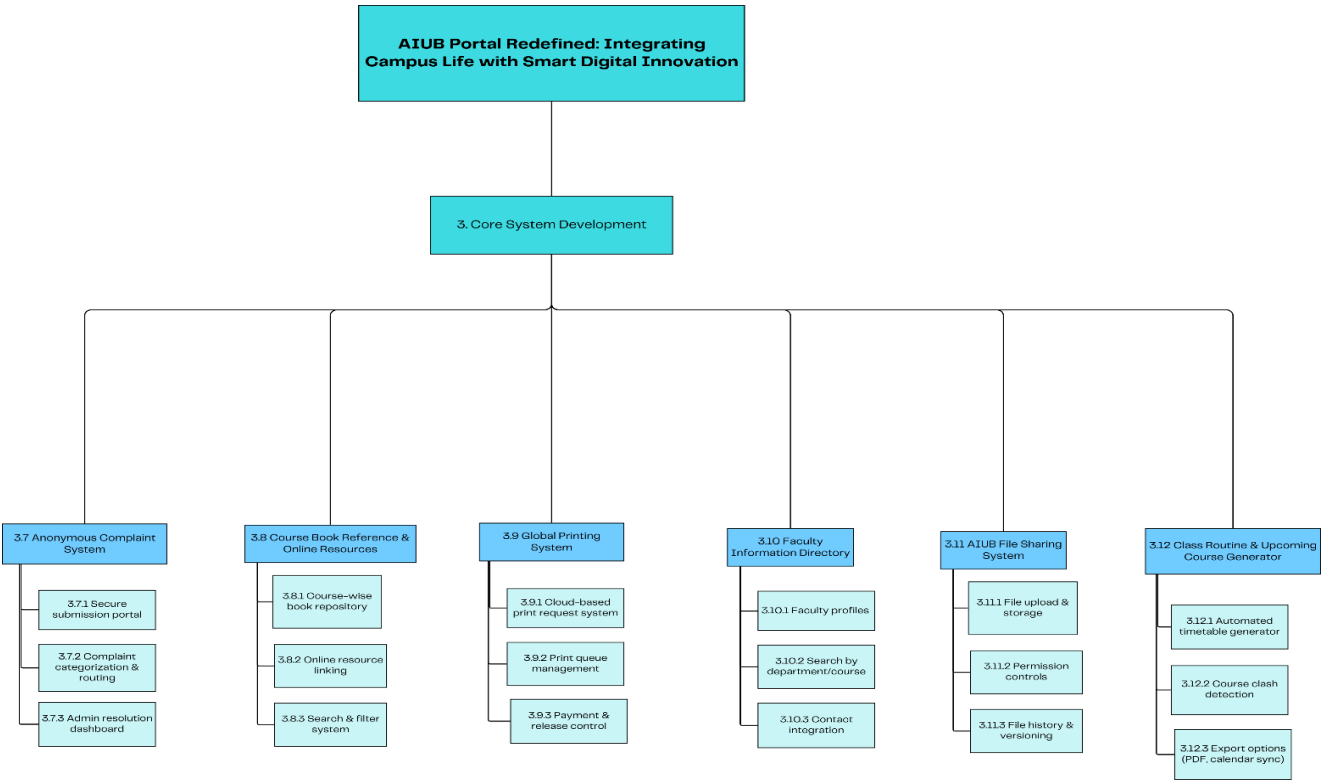
$$= 1.25 \approx 2 \text{ Persons}$$

3.2 Project Scheduling

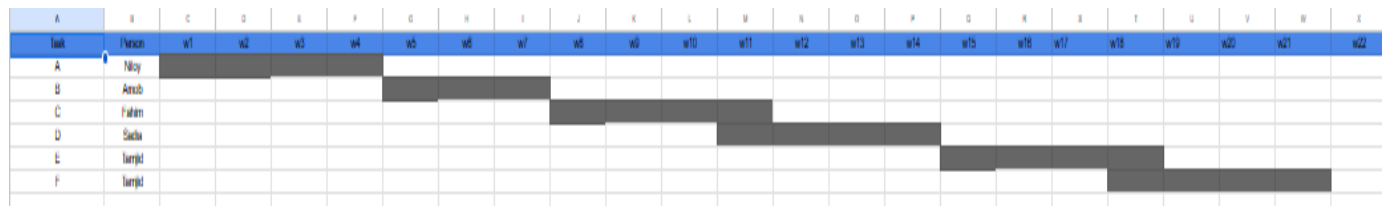
3.2.1 Work Breakdown Structure (WBS):



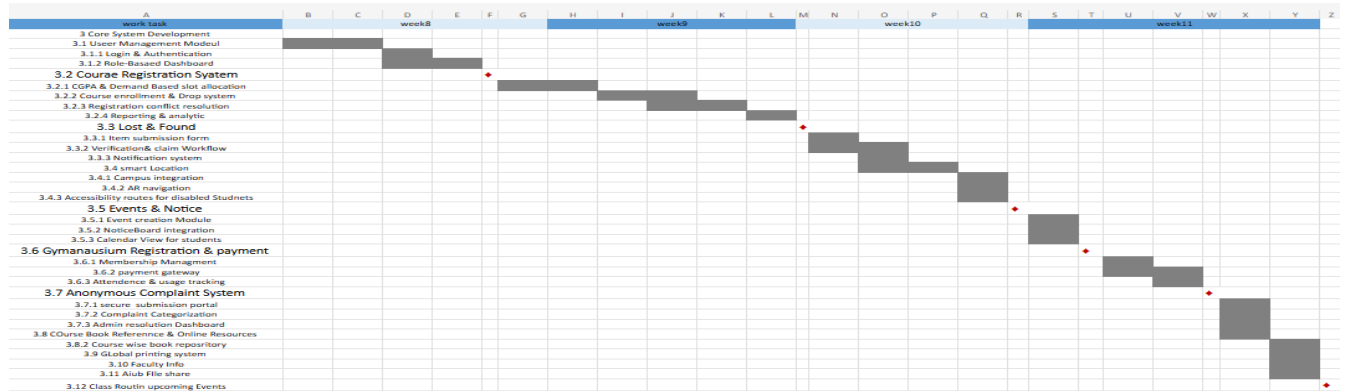
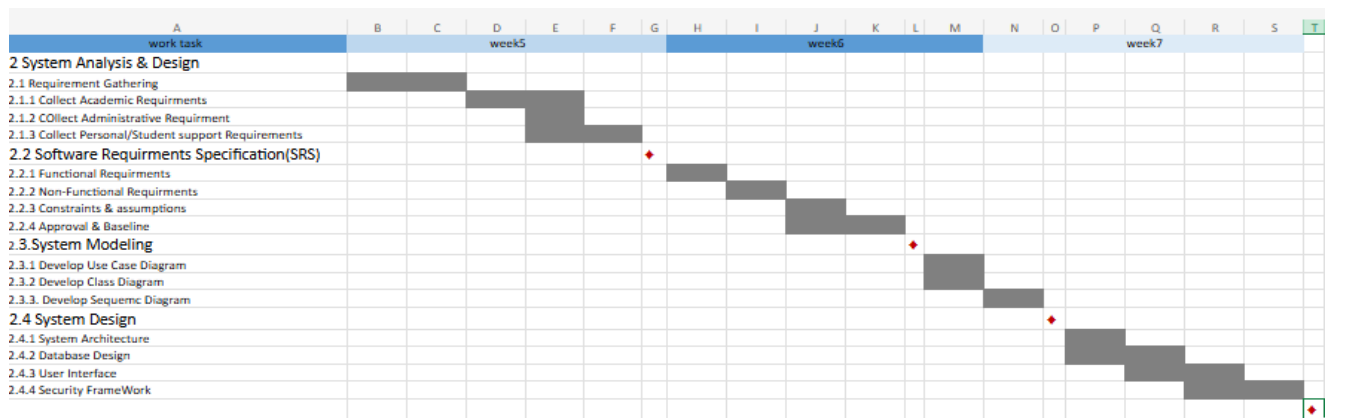
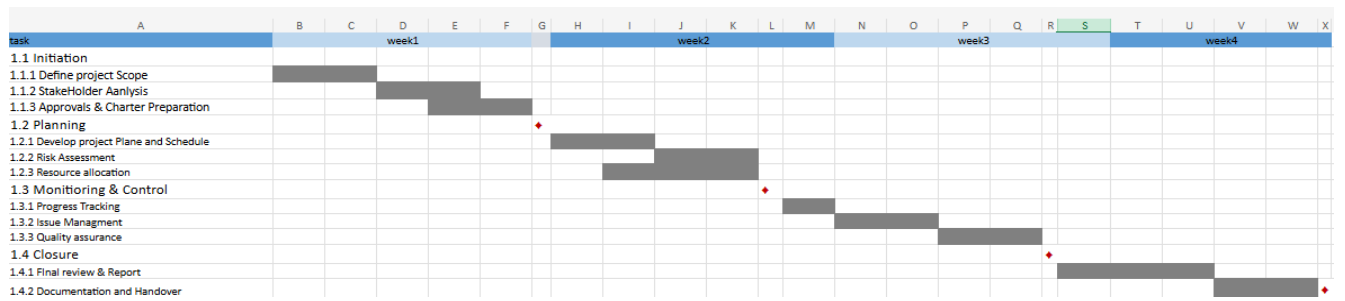
3.2.I Work Breakdown Structure (WBS) Continued:



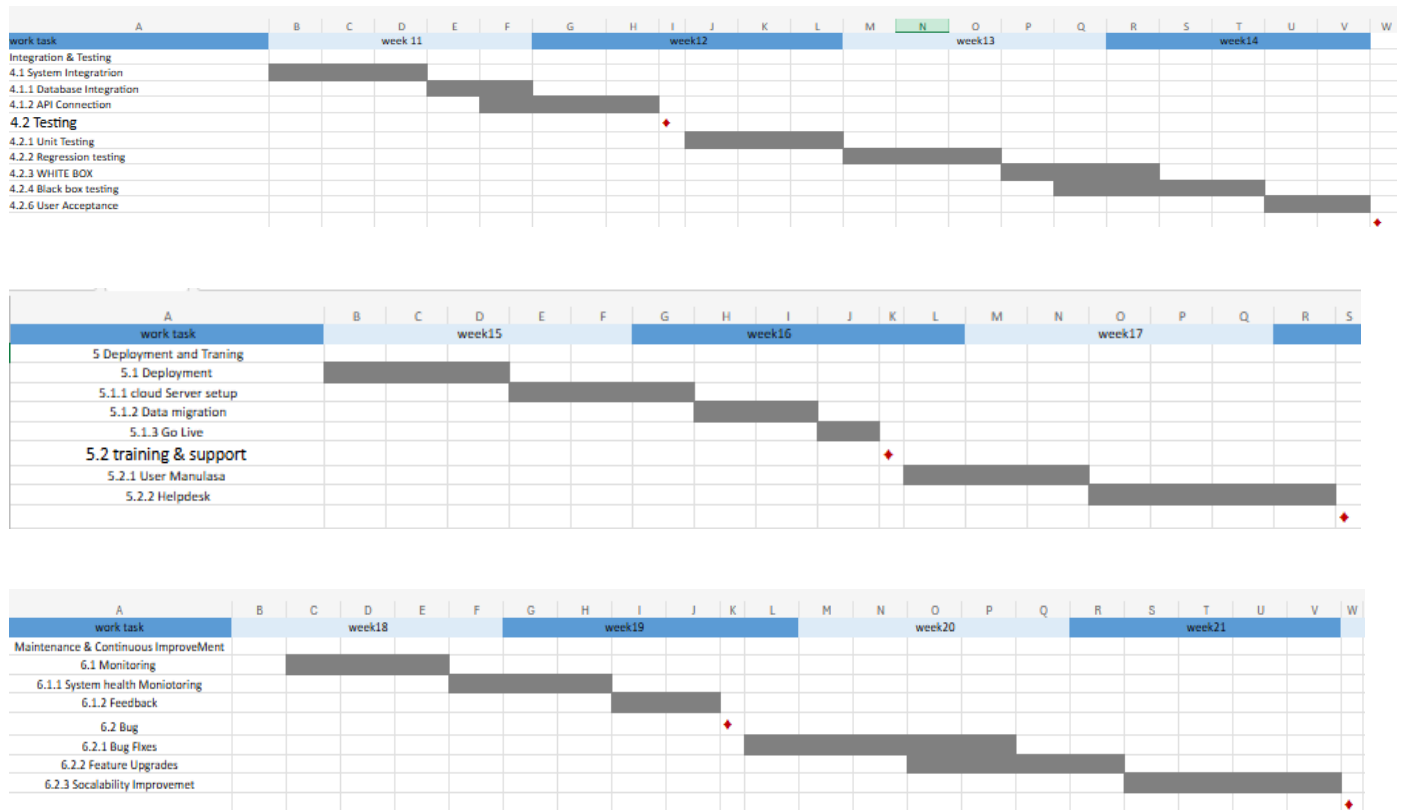
3.2.II Overview Timeline:



3.2.III Detailed Timeline:



3.2.III Detailed Timeline:



3.2.IV Allocate Effort (40–20–40 Guideline)

We'll distribute effort across **Analysis/Design, Coding, and Testing**:

- **System Analysis & Design (40%)**
 - Requirements gathering, modeling, UI/UX design.
 - Deliverables: Requirement Document, System Architecture, Database Schema.
- **Coding/Development (20%)**
 - Core system development (all modules).
 - Deliverables: Completed modules (User Mgmt., Course Reg., Payment, etc.).
- **Testing (40%)**
 - Unit testing, integration testing, user acceptance testing.
 - Deliverables: Test Reports, Bug Fix Logs, User Acceptance Sign-off.

3.2.V Earned Value Analysis (EVA)

Our project budget is \$50,000 total, spread across 20 weeks.

So, the average planned value per week (BCWS) = $\frac{50000}{20} = \$2500$ *Per week*

Now let's check progress in **Week 10 (mid-point)**:

- **Planned Value (BCWS):**

At week 10 $\rightarrow 10 \times 2500 = \25000

- **Earned Value (BCWP):**

Assuming, only 85% of planned work is done by week 10

$25000 \times 0.85 = \$21250$

- **Actual Cost (ACWP):**

Our costs have reached \$30,000 by week 10.

- Schedule performance index, $SPI = BCWP/BCWS$

$$= \$21250/\$25000 = 0.85 < 1$$

Behind schedule (only 85% of planned progress achieved).

- Schedule variance, $SV = BCWP - BCWS$

$$= \$21250 - \$25000 = -\$3750$$

Behind schedule by \$3750 worth of work.

- Cost performance index, $CPI = BCWP/ACWP$

$$= \$21250/\$30000 = 0.708 < 1$$

Cost efficiency is poor (spending more than planned).

- Cost variance, $CV = BCWP - ACWP$

$$= \$21250 - \$30000 = -\$8750$$

Over budget by \$8750.

3.2.VI Project Risk Management

Risk management for the AIUB Portal project entails identifying potential uncertainties, assessing their likelihood and impact, and developing strategies to mitigate them. This ensures successful integration of campus features like authentication, gym registration, cloud printing, lost & found, e-books, faculty directory, anonymous complaints, file sharing, notices, academic registration, and location mapping. Using an agile process, risks are monitored through sprints, with regular reviews to adjust plans, minimizing disruptions to timeline, budget, and quality while addressing societal and environmental sustainability.

Risk Categories:

- PS (Product Size): Risks tied to software scope, requirements, and overall size.
- BU (Business Impact): Risks from management, market, or budget constraints.
- CU (Customer Characteristics): Risks involving user sophistication and communication.
- PR (Process Definition): Risks related to software process adherence and definition.
- DE (Development Environment): Risks in tools, facilities, and development support.
- TE (Technology to be Built): Risks from system complexity and new technologies.
- ST (Staff Size and Experience): Risks due to team skills, experience, and availability.

Impact values:

1. Catastrophic
2. Critical
3. Marginal
4. Negligible

RMMM:

The work product is called Risk Mitigation, Monitoring, and Management Plan.
RMMM strategies derived from checklists and Boehm's top risks.

Risk Management Table

Risk	Category	Probability	Impact	RMMM
Feature scope expansion beyond initial estimates	PS	65%	2	Mitigate: Define clear scope in SRS; use agile backlog prioritization. Monitor: Track requirements changes in sprint reviews. Manage: Implement changing control board for approvals.

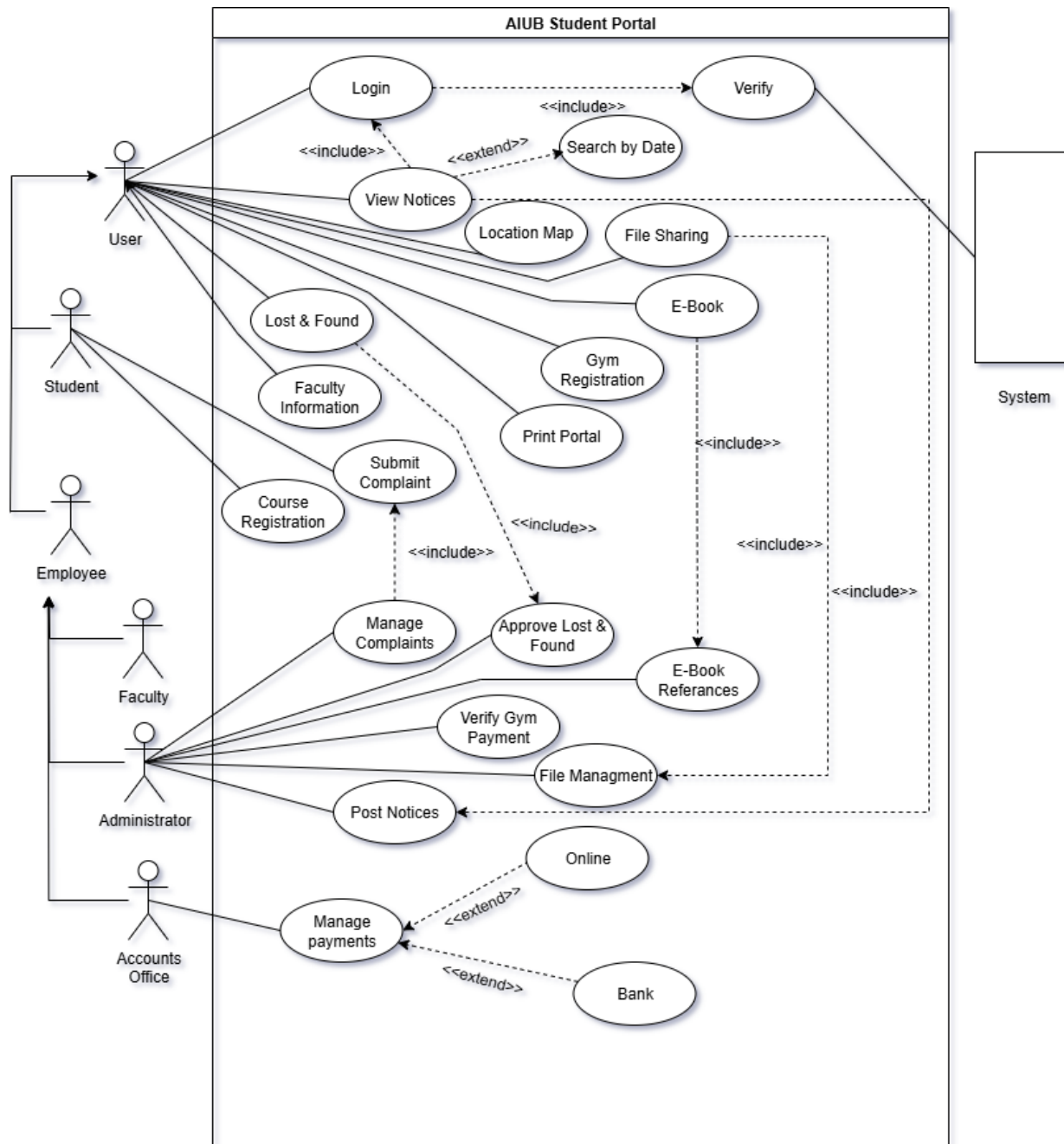
Risk	Category	Probability	Impact	RMMM
Unexpected surge in concurrent users during peak times	PS	35%	3	Mitigate: Design scalable architecture with cloud resources. Monitor: Use performance metrics dashboards. Manage: Conduct stress tests and auto-scale setups.
Limited reuse of existing campus APIs	PS	55%	2	Mitigate: Assess APIs early via prototypes. Monitor: Review of integration progress weekly. Manage: Develop custom wrappers as backups.
User reluctance to adopt new portal features	BU	45%	3	Mitigate: Run beta testing with feedback loops. Monitor: Track usage analytics post-launch. Manage: Offer incentives and tutorials for adoption.
Shortened project timeline due to academic calendar	BU	60%	2	Mitigate: Build buffers in Gantt chart. Monitor: Weekly progress meetings. Manage: Reprioritize non-core features if needed.
Loss of stakeholder support or funding	CU	30%	1	Mitigate: Regular demos to stakeholders. Monitor: Feedback surveys. Manage: Secure alternative resources early.
Frequent requirement changes from university admins	CU	75%	2	Mitigate: Use formal requirement elicitation. Monitor: Change logs. Manage: Agile iterations for quick adaptations.
New tech like real-time location failing expectations	TE	40%	1	Mitigate: Proof-of-concept trials. Monitor: Tech evaluations. Manage: Fallback to simpler GPS methods.
Insufficient training on Figma or dev tools	DE	70%	3	Mitigate: Schedule workshops. Monitor: Skill assessments. Manage: Pair

Risk	Category	Probability	Impact	RMMM
				junior with senior developers.
Team members lacking experience in web security	ST	50%	2	Mitigate: External training sessions. Monitor: Code reviews. Manage: Consult experts for critical modules.
High team turnover mid-project	ST	40%	2	Mitigate: Foster team building. Monitor: Retention check-ins. Manage: Cross-training and documentation.
Integration failures between modules like printing and payments	TE	60%	2	Mitigate: Incremental integration testing. Monitor: CI/CD pipelines. Manage: Dedicated integration sprints.
Data privacy breaches in file sharing or complaints	TE	45%	1	Mitigate: Implement GDPR-compliant security. Monitor: Security audits. Manage: Incident response plan
Performance lags in e-book downloads during high demand	PR	50%	3	Mitigate: Optimize with caching. Monitor: Load monitoring tools. Manage: Throttle access if overloaded
Inaccurate faculty directory updates	CU	55%	3	Mitigate: Automated sync with HR systems. Monitor: Data validation checks. Manage: Manual review protocols
Gym slot overbooking due to concurrency issues	PS	30%	2	Mitigate: Use locking mechanisms in database. Monitor: Transaction logs. Manage: Alert system for conflicts.

4. SOFTWARE DESIGN

4.1 System Design

4.1.1 Use-Case Diagram

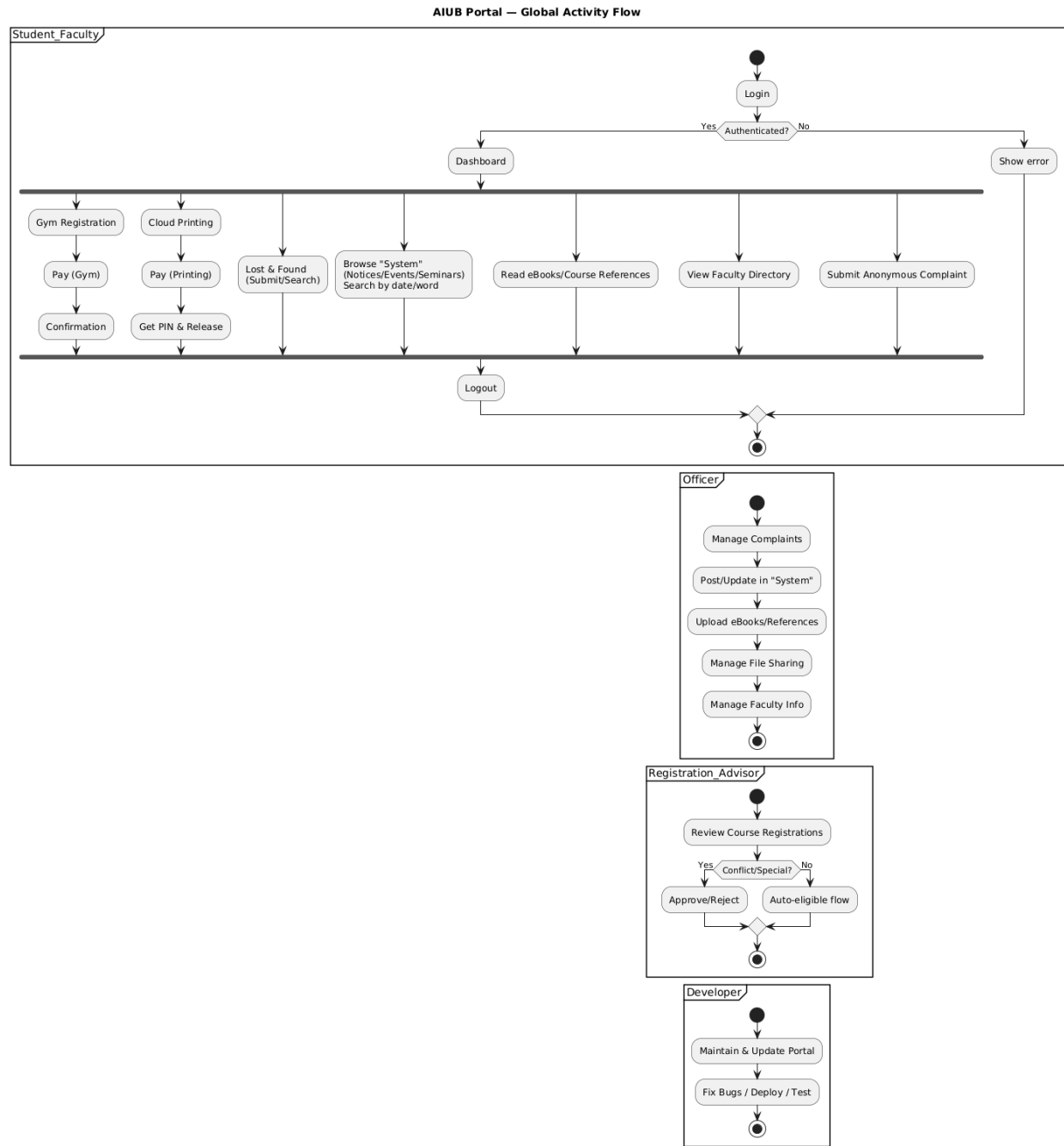


```

classDiagram
    class User {
        +userid: int
        +name: String
        +email: String
        +password: String
        +login(): bool
        +logout(): void
    }
    class Administrator {
        +adminId: int
        +role: String
        +approveLostItem(temId: int): void
        +manageComplaint(complaintId: int): void
        +postNotice(notice: Notice): void
    }
    class Faculty {
        +facultyId: int
        +department: String
        +designation: String
        +uploadMaterial(file: File): void
        +updateFacultyInfo(): void
    }
    class Student {
        +studentId: int
        +program: String
        +cpga: float
        +registerCourse(courseId: int): bool
        +submitComplaint(text: String): void
        +submitLostItem(desc: String): void
        +payFee(amount: float): bool
    }
    class Course {
        +courseId: int
        +title: String
        +schedule: String
        +capacity: int
        +checkAvailability(): bool
        +waitlistStudent(studentId: int): void
    }
    class EBook {
        +bookId: int
        +title: String
        +author: String
        +courseId: int
        +download(): File
        +viewReference(): List<String>
    }
    class Notice {
        +noticeId: int
        +title: String
        +content: String
        +date: Date
        +category: String
        +postNotice(): void
        +filterNotice(type: String): List<Notice>
    }
    class FileSharing {
        +field: int
        +fileName: String
        +uploadId: int
        +expiryDate: Date
        +secureLink: String
        +uploadFile(file: File): void
        +shareLink(): String
        +autoDelete(): void
    }
    class FacultyInformation {
        +facultyId: int
        +name: String
        +department: String
        +officeLocation: String
        +contact: String
        +searchFaculty(name: String): FacultyInformation
        +updateFacultyInfo(): void
    }
    class LocationMap {
        +mapId: int
        +buildingName: String
        +roomNumber: String
        +facultyLocation: String
        +searchBuilding(name: String): Location
        +navigate(from: String, to: String): Path
        +filterByType(type: String): List<Location>
    }
    class GymRegistration {
        +regId: int
        +studentId: int
        +serviceType: String
        +paymentStatus: String
        +enrollGym(studentId: int): void
        +verifyPayment(): bool
    }
    class PrintPortal {
        +printId: int
        +studentId: int
        +fileName: String
        +status: String
        +uploadDoc(file: File): void
        +sendToPrinter(): void
        +notifyReady(): void
    }
    class AccountsOffice {
        +officeId: int
        +officeName: String
        +verifyPayment(paymentId: int): bool
        +trackPayment(studentId: int): String
    }
    class Payment {
        +paymentId: int
        +amount: float
        +status: String
        +date: Date
        +verifyPayment(): bool
        +trackPayment(): String
    }
    class Complaint {
        +complaintId: int
        +content: String
        +isAnonymous: bool
        +status: String
        +submit(): void
        +resolve(): void
    }
    class LostAndFoundItem {
        +itemId: int
        +description: String
        +status: String
        +submittedBy: int
        +submitItem(): void
        +approveItem(): void
    }

    User <|-- Administrator
    User <|-- Faculty
    User <|-- Student
    Administrator "1" -- "0..*" Notice : Posts
    Administrator "1" -- "0..*" FileSharing : manages
    Faculty "1" -- "0..*" FacultyInformation : creates
    Faculty "1" -- "0..*" LocationMap : profile
    Faculty "1" -- "0..*" LocationMap : locates
    Student "0..*" -- "1" Course : enrolls
    Student "1" -- "0..*" EBook : references
    Student "1" -- "0..*" PrintPortal : prints
    Student "1" -- "0..*" GymRegistration : registers
    Student "1" -- "0..*" GymRegistration : owns
    Student "0..*" -- "0..*" Complaint : submits
    Student "1" -- "0..*" Payment : pays
    Student "1" -- "0..*" AccountsOffice : verifies
    Course "0..*" -- "1" EBook : references
    Notice "0..*" -- "0..*" FileSharing : creates
    FileSharing "0..*" -- "0..*" Complaint : creates
    FacultyInformation "0..*" -- "0..*" Complaint : creates
    LocationMap "0..*" -- "0..*" Complaint : creates
    GymRegistration "0..*" -- "0..*" Complaint : creates
    PrintPortal "0..*" -- "0..*" Complaint : creates
    AccountsOffice "0..*" -- "0..*" Complaint : creates
    Payment "0..*" -- "0..*" Complaint : creates
    Complaint "0..*" -- "0..*" LostAndFoundItem : submits
    Complaint "0..*" -- "0..*" Payment : pays
    Complaint "0..*" -- "0..*" AccountsOffice : verifies
    Complaint "0..*" -- "0..*" Payment : trackPayment
    Complaint "0..*" -- "0..*" AccountsOffice : trackPayment
    
```


4.1.3 Activity Diagram



4.1.4 Sequence Diagram

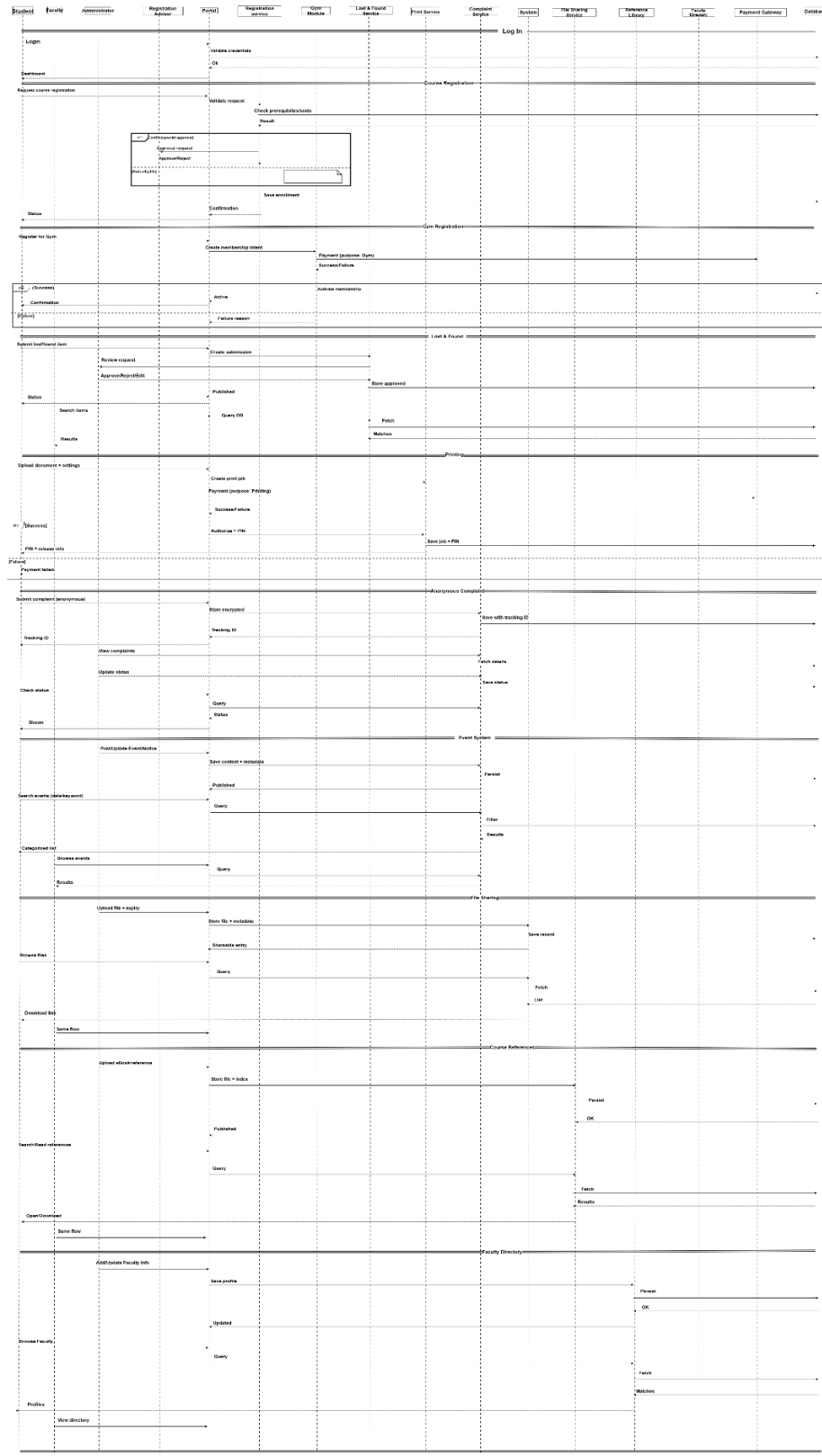
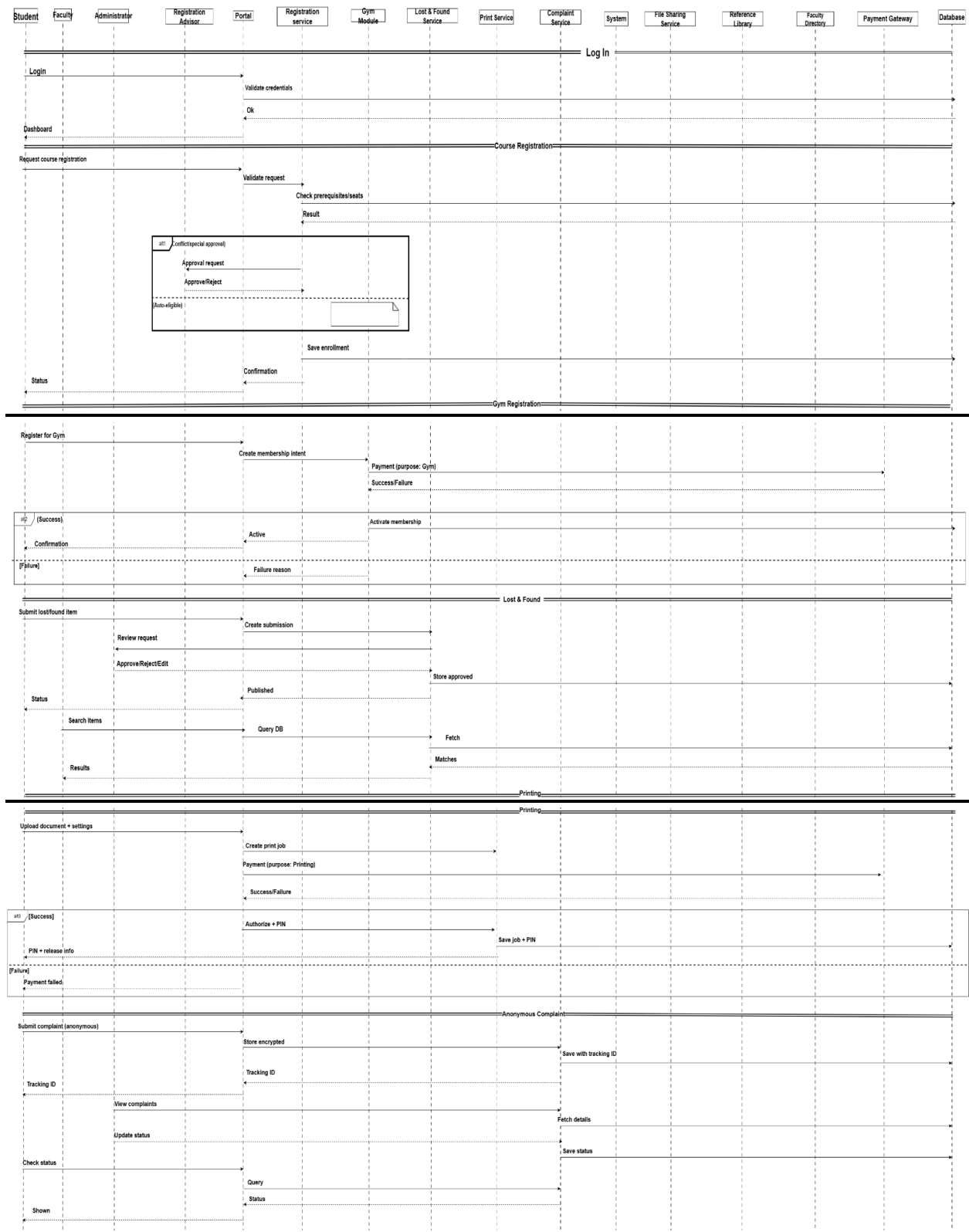
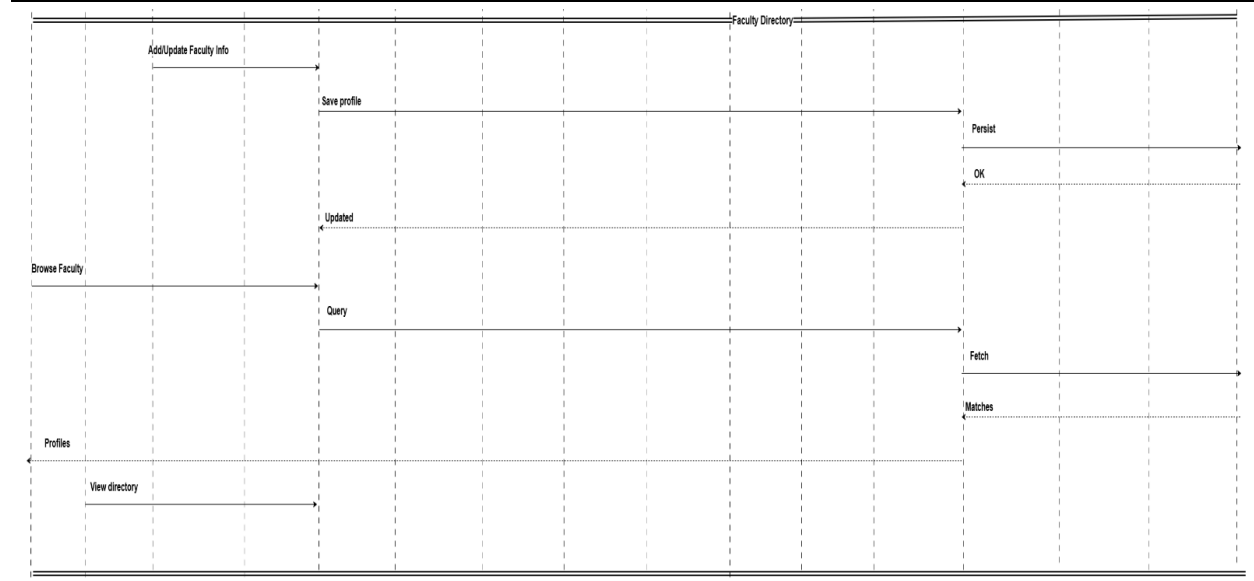
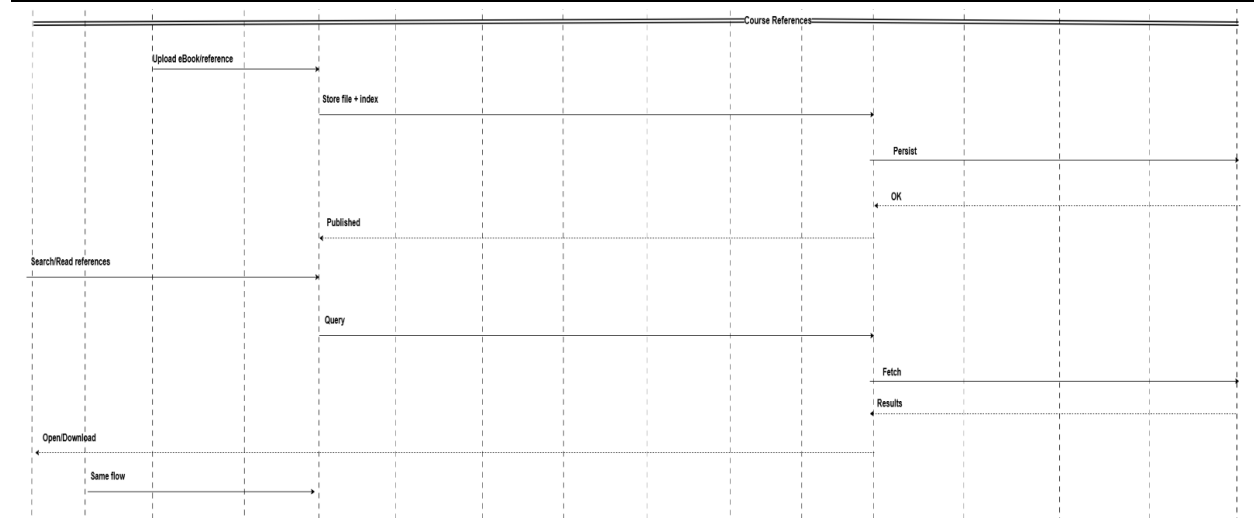
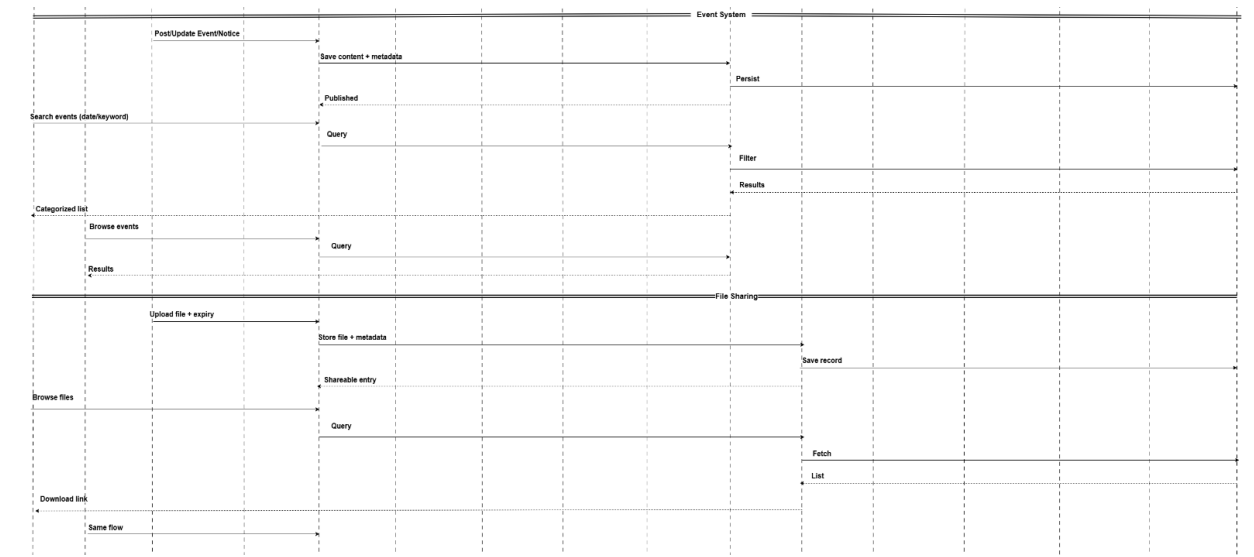


Figure: Sequence Diagram Full Part given above and Divided Part given below





4.2 UI / Wireframe Design

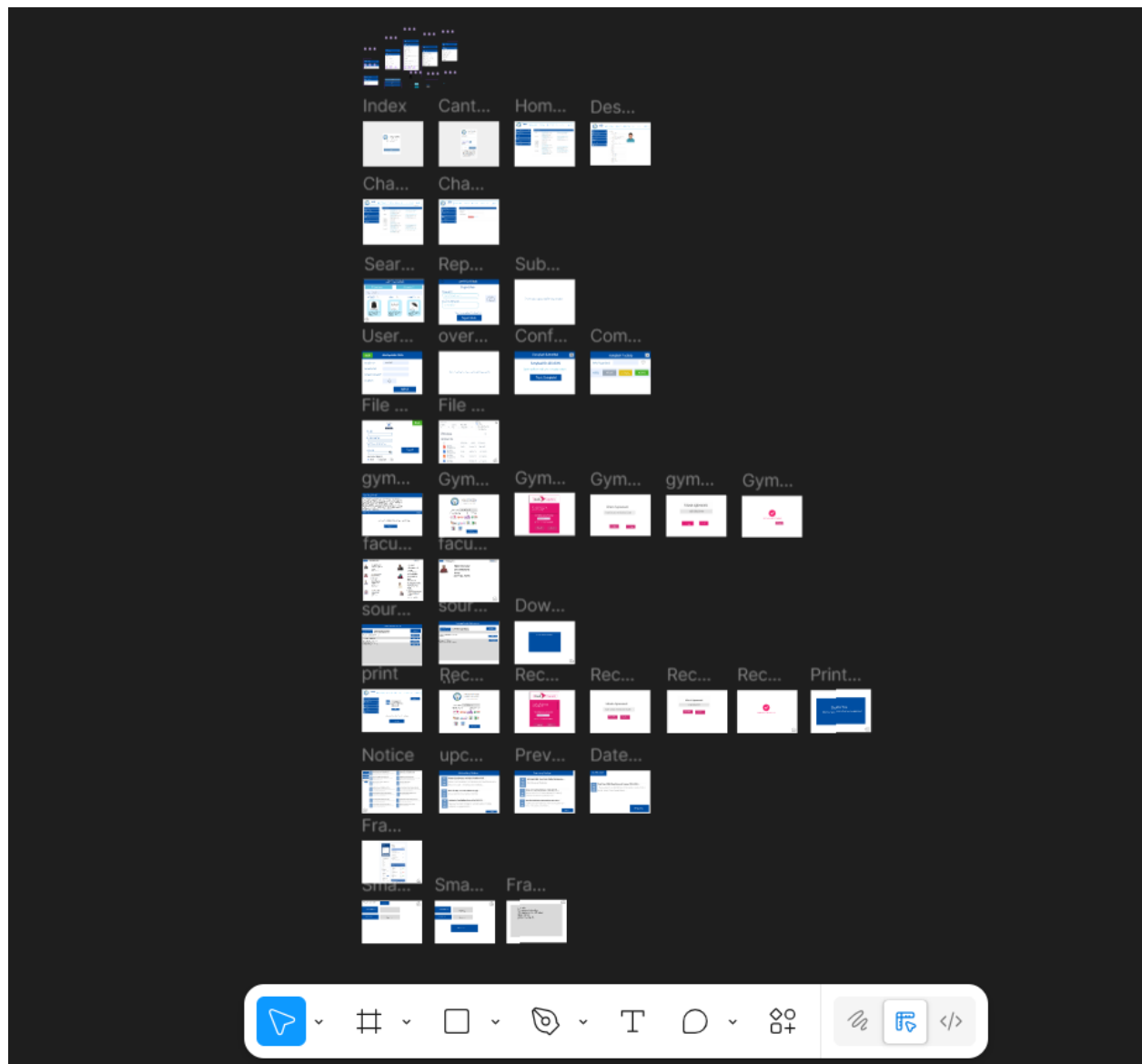


Figure: Whole Figma Design

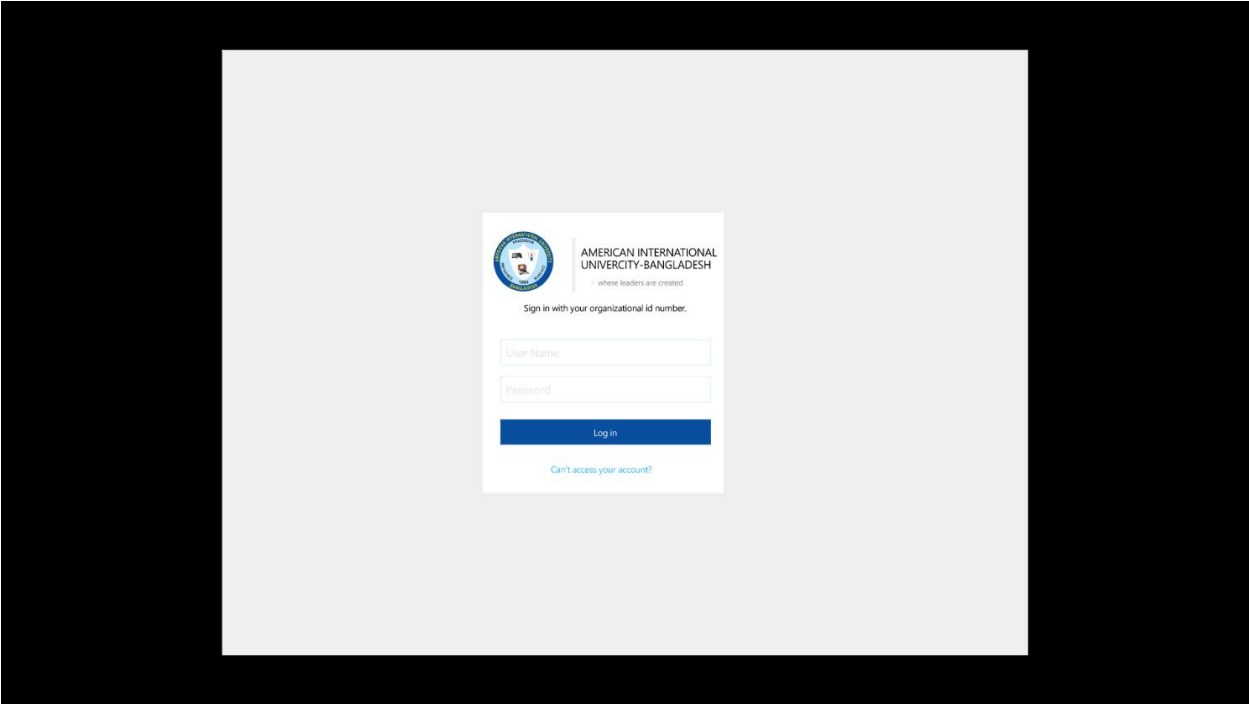


Figure: Login Page

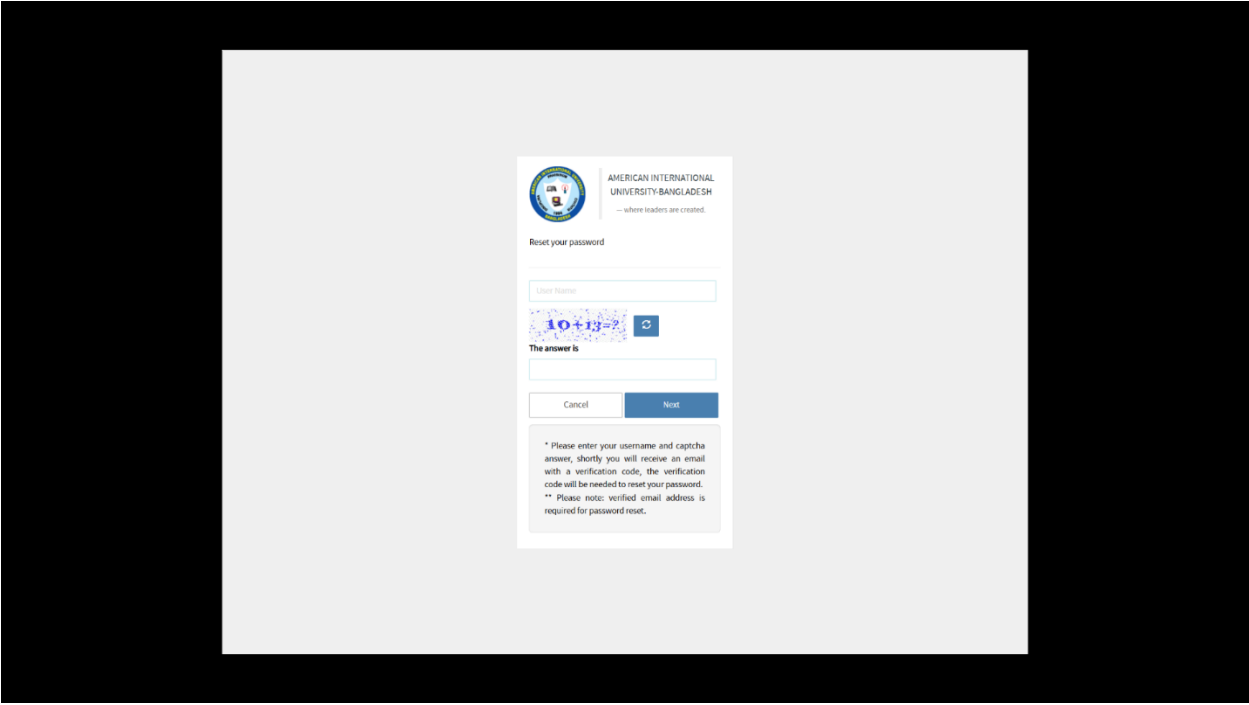


Figure: Reset Password


AIUB
 PORTAL

[Course & Result](#)
[Registration](#)
[Grade Report](#)
 Welcome PAUL, NILOY

 Academics
  Grade Reports
  Library
  Others
  Messages
  Services

Class Schedule

Date	Course	Time	Room
Today	SOFTWARE ENGINEERING [B]	Tue 2:00 PM - Tue 5:00 PM	D50103
	MICROPROCESSOR AND EMBEDDED SYSTEMS [B]	Tue 11:00 - Tue 2:00 PM	DN0310
Tomorrow	COMPIER DESIGN [B]	Tue 8:00 - Tue 11:00	D50107
	MICROPROCESSOR AND EMBEDDED SYSTEMS [B]	Wed 11:20 - Wed 12:50 PM	9401
28-Aug-2025	DATA COMMUNICATION [I]	Wed 1:00 PM - Wed 2:30 PM	9311
29-Aug-2025	No Class On This Day		
30-Aug-2025	No Class On This Day		
31-Aug-2025	DATA COMMUNICATION [I]	Sun 11:00 - Sun 2:00 PM	D50108
01-Sep-2025	SOFTWARE ENGINEERING [B]	Mon 2:40 PM - Mon 4:10 PM	3208
02-Sep-2025	COMPIER DESIGN [B]	Mon 8:00 - Mon 9:30	9409
03-Sep-2025	SOFTWARE ENGINEERING [B]	Tue 2:00 PM - Tue 5:00 PM	D50103
04-Sep-2025	COMPIER DESIGN [B]	Tue 8:00 - Tue 11:00	D50107

Figure: Home Page


AIUB
 PORTAL

[Course & Result](#)
[Registration](#)
[Grade Report](#)
 Welcome PAUL, NILOY

 Academics
  Grade Reports
  Library
  Others
  Messages
  Services

PAUL, NILOY

Student ID:	20-61775-2
CGPA:	
Credit:	00
Program:	Bachelor of Science in Computer Science & Engineering
Department:	INSTITUTE OF SCIENCE & TECHNOLOGY
Core:	BS-CSE, Summer 17-18, Core
Major:	
Second Major:	
Minor:	
Elective:	BS-CSE, Summer 17-18, Elective
Father Name:	
Mother Name:	
Parent Address:	
Permanent Address:	
Verified Contact:	
Verified Email:	
DOB:	05/11/2004
Sex:	Male
Nationality:	Bangladeshi
Religion:	Hindu
Marital Status:	Single
Blood Group:	O+
Admission Date:	12-Apr-2023
Graduation Date:	



Figure: Profile Page

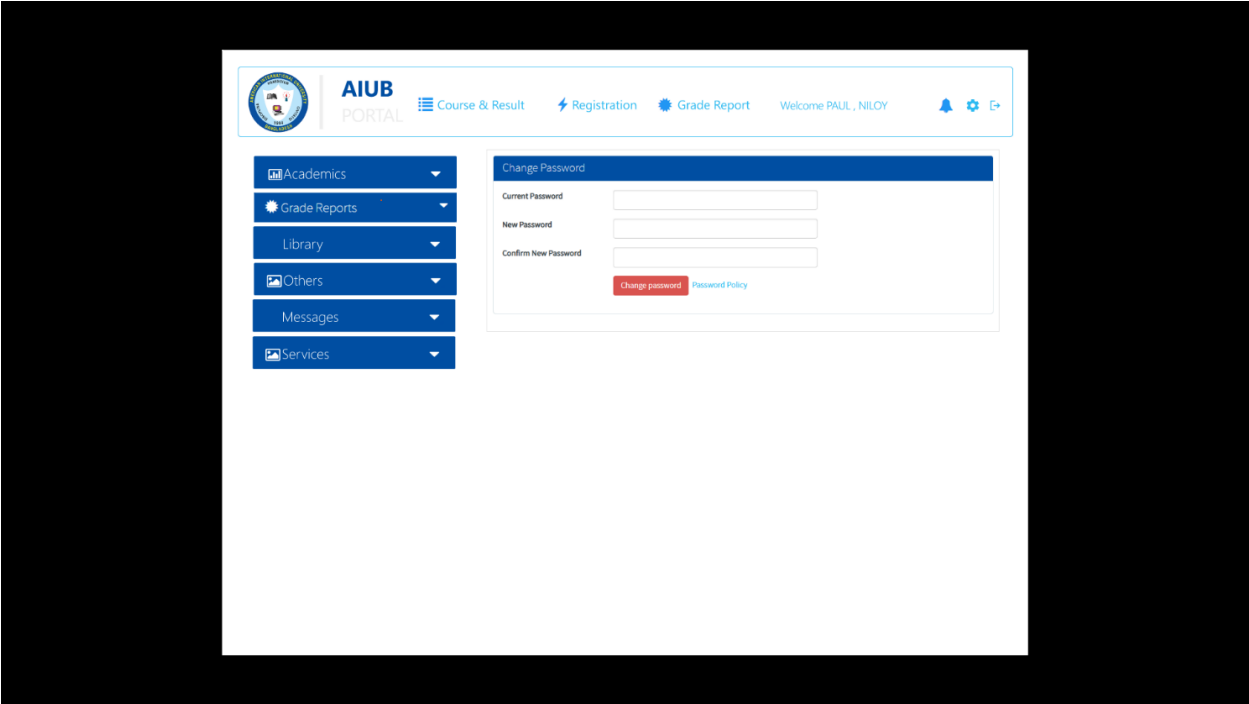


Figure: Change Password

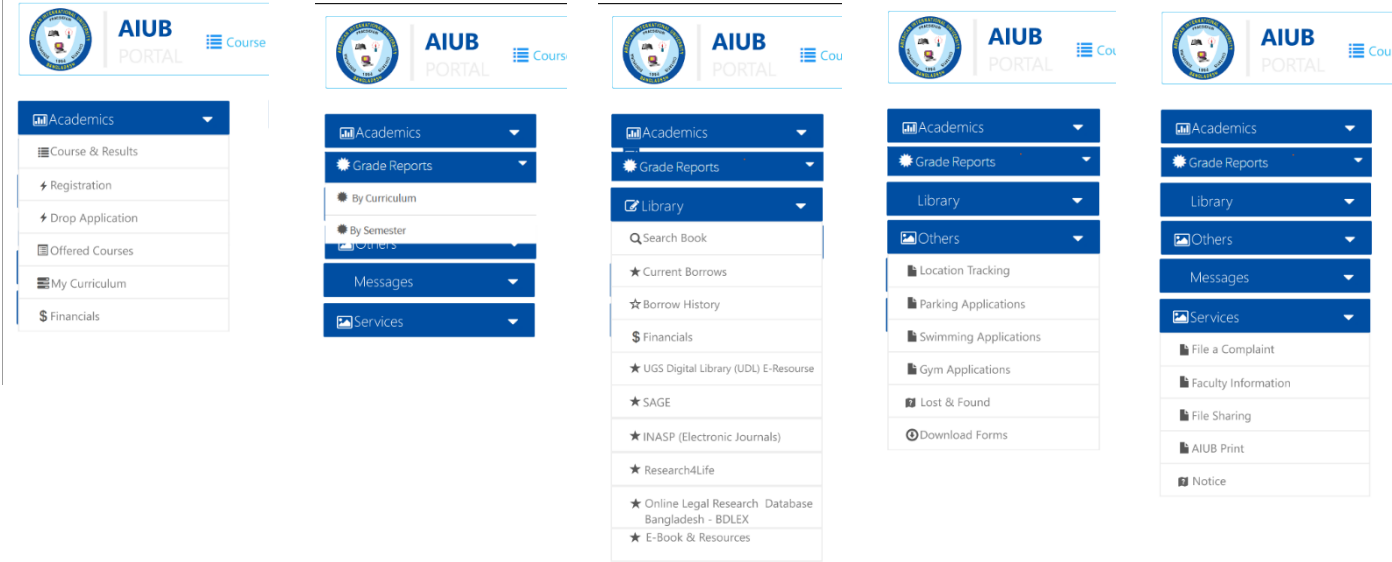


Figure: Drag & Drop Portion

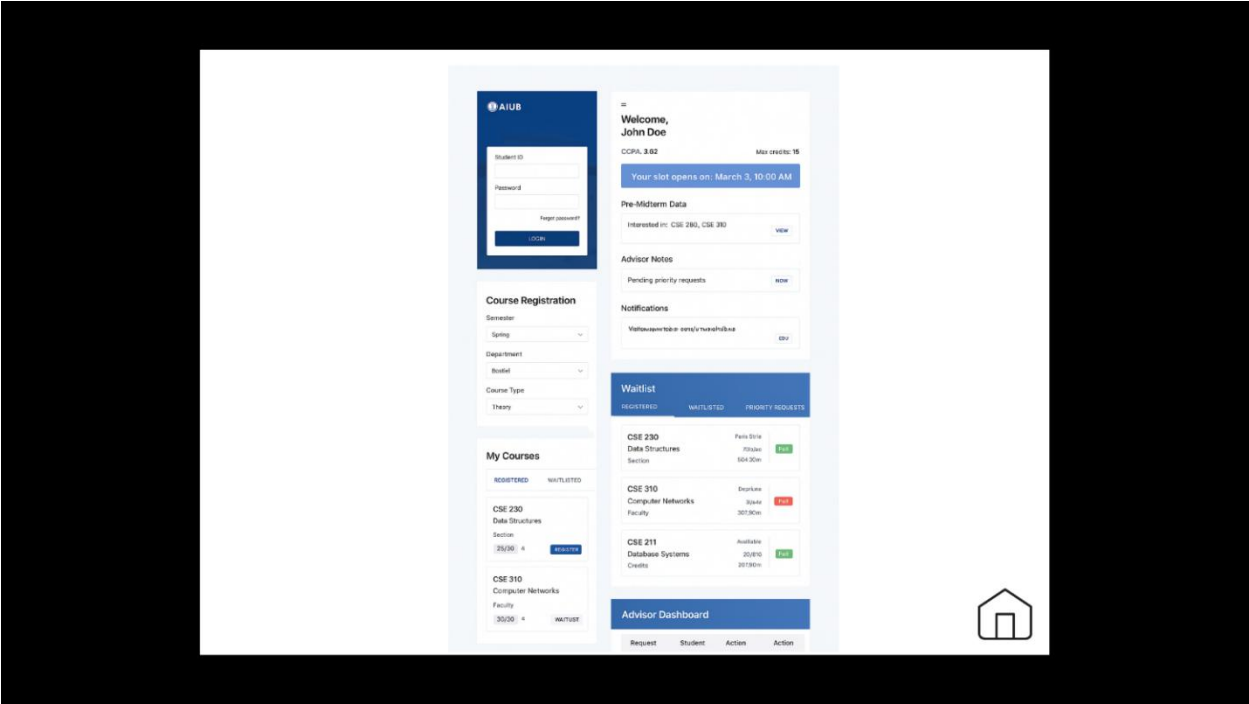


Figure: Registration Process

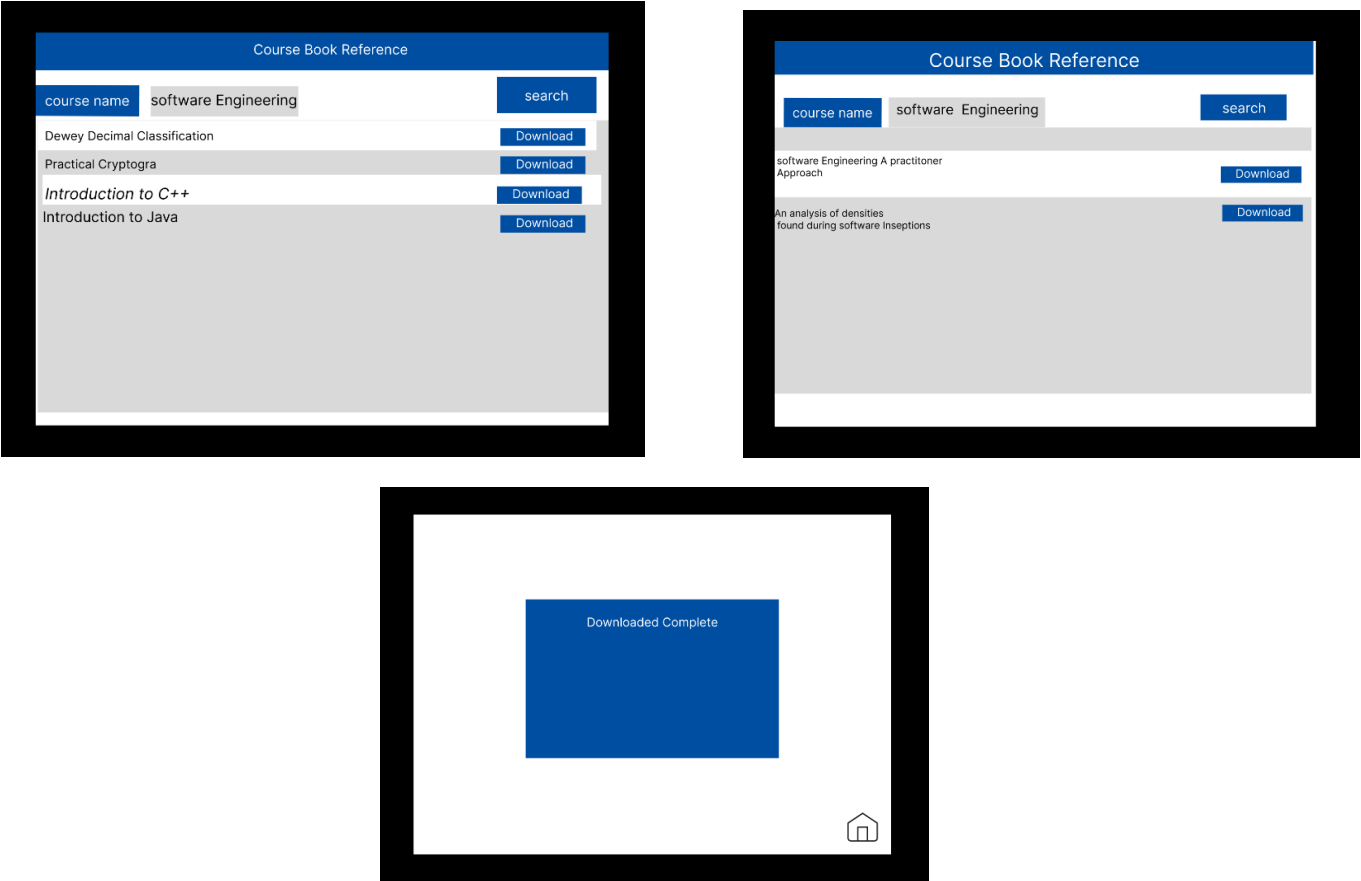


Figure: E-Book

(Turn on your location)

On

current location

destination

Annex 6

current location

D Building

destination

Annex 6

Located me

Turn left.

Go towards C Building.

At C Building, turn left again.

Walk straight.

Arrive at Annex 6.

Figure: Location System

LOST & FOUND

Report Item

Description

Enter item description

Location found/lost

Enter location

Admin approval required for public visibility

Report Item

LOST & FOUND

Report Item

Search Item

Q Search...

Category

Date

Location

Black Backpack

Lost near the library

August 15, 2025

Campus

Eyeglasses

Lost near the Cafeteria

August 02, 2025

Campus

Umbrella

Lost near the library

June 23, 2025

Campus

Figure: Lost & Found

Time Schedule

For ensuring proper fitness of all students and employees, AIUB possesses a highly facilitated with modern fitness equipment gymnasium which is accessible for both male and female students, faculty and officials. The gym is located at the 10th floor of D-building of AIUB permanent campus. There are two instructors who help the beginners with the workout routine and training.

There are total three sessions available to avail the gym facilities out of which one session dedicated only for female students (from 9.00 am-12.00 pm) and one for male students (from 12.00 pm-4.30 pm). There is a session (from 5.00 pm) specially for faculties and officials.

Active Status :

Disable

(click Registration Button to Enable Your Gymnasium membership)

Registration



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

— where leaders are created.

User Name

22-48593-3

Mobile Banking

(please select one)














Continue

bKash

Payment

Merchant: Aiub Accounts

Invoice No: 1230348334

Amount : BDT 350

Your bKash account number

01700000001

☒ I agree to the terms and conditions

proceed

cancel

bKash Agreement

Enter bKash Verification Code

proceed

Cancel

bKash Agreement

Enter bKash PIN

proceed

cancel



Agreement Creation Successful

Continue

Figure: Gymnasium Maintenance

Back

Anonymous Form

Complaint ID

Auto Generated

Complaint Type

Complaint Description

File Upload

+

Submit

Complaint Submitted

Complaint ID: AB123456

Save this ID to track your complaint status.

Track Complaint

Your complaint has been submitted anonymously.

Complaint Tracking

Enter Complaint ID

Status

Pending

In Review

Resolved

Figure: Anonymous Complaint

name TONNY SHEKHA KAR search



MD. HASIBUL HASAN
hasib.hasan@aiub.edu
Faculty
Room No:DS041



MD. RIFAT
rifat.m@aiub.edu
Faculty
Room No:DS0412



NAZIA ZAMAN CHOWDHURY
naziazaman@aiub.edu
Faculty
Room No:DS043



DR. SALAIKA PARVIN
skparvin@aiub.edu
Faculty
Room No:DS044



MAHFUJUR RAHMAN
mahfuj@aiub.edu
Faculty
Room No:DS045



MD. MAHMUDUL HASAN
mahmud@aiub.edu
Faculty
Room No:DS046



MD. SHOFIQUIL ISLAM
shofiq74@aiub.edu
Faculty
Room No:DS047



DR. M. MOSTAFIZUR RAHMAN
mostafiz.math@aiub.edu
Faculty
Room No:DS048

name TONNY SHEKHA KAR search



TONNY SHEKHA KAR
tonny.kar@aiub.edu
Faculty
Room No : 9408J



Figure: Faculty Details

Back

Drag & Drop



or

Upload from Device

File Title

Description(optional)

Select Course / Group / Faculty

Expiry Date



Permission Settings

☒ View

☐ Download

☐ Edit

Upload

Figure: File Sharing System

10 /8/2025	
10 Aug 2025	Seat Plan of Mid-Term Exams of Summer 2024-2025 ... Please contact Exam Control Room if your name or course is not in the list Annex 2 Exam Control Room # ...

Previous Notice	
13 Aug 2025	BBA Set B Mid-Term Exam Policy for Summer ... Click here to see the Notice
12 Aug 2025	Notice for Freshman Students of FALL 2025-26 ... Attention: Notice for All Freshman Students (Fall : 2025-26) *Freshman students who got admitted in Fall ...
11 Aug 2025	Seat Plan of Mid-Term Exams of Summer 2024-2025 ... Please contact Exam Control Room if your name or course is not in the list Annex 2 Exam Control Room # ...

Upcoming Notice	
25 Aug 2025	Written Result Notice and Viva Schedule of Fall ... RESULT & VIVA SCHEDULE OF ADMISSION WRITTEN EXAM HELD ON AUGUST 21, 2025 On-Campus Viva of selected ...
25 Aug 2025	Notice for EEE, CoE & IPE Students to apply ... Apply for Capstone Project Apply for Internship
20 Aug 2025	Admission Test Written Exam of Fall 2025-26 ... Admission Test WRITTEN RESULT Fall 2025-26, (Slot 1) will be published on August 25, 2025 ...

Figure: Searching Notice, Upcoming, Previous and Current Program Notice

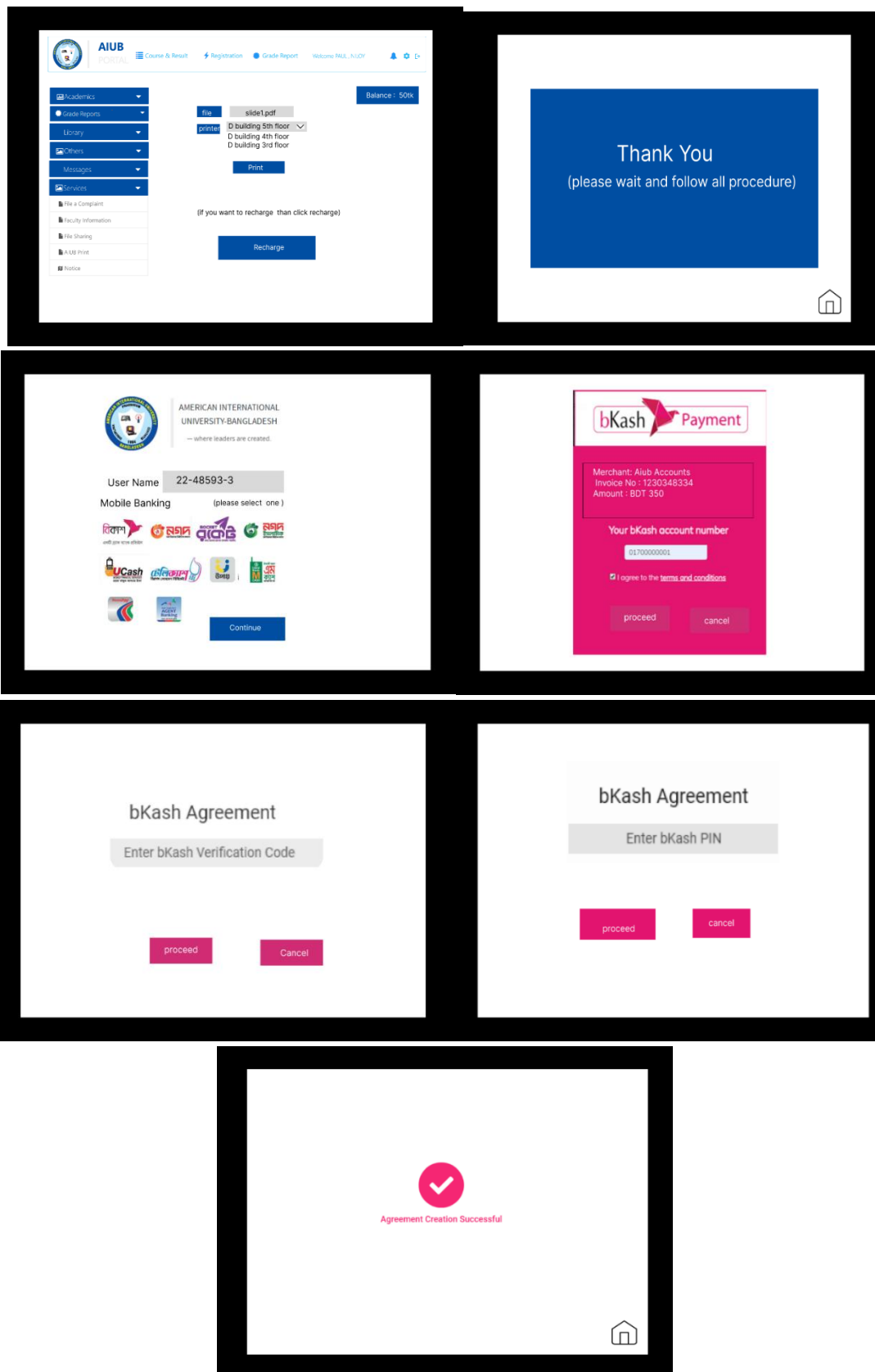
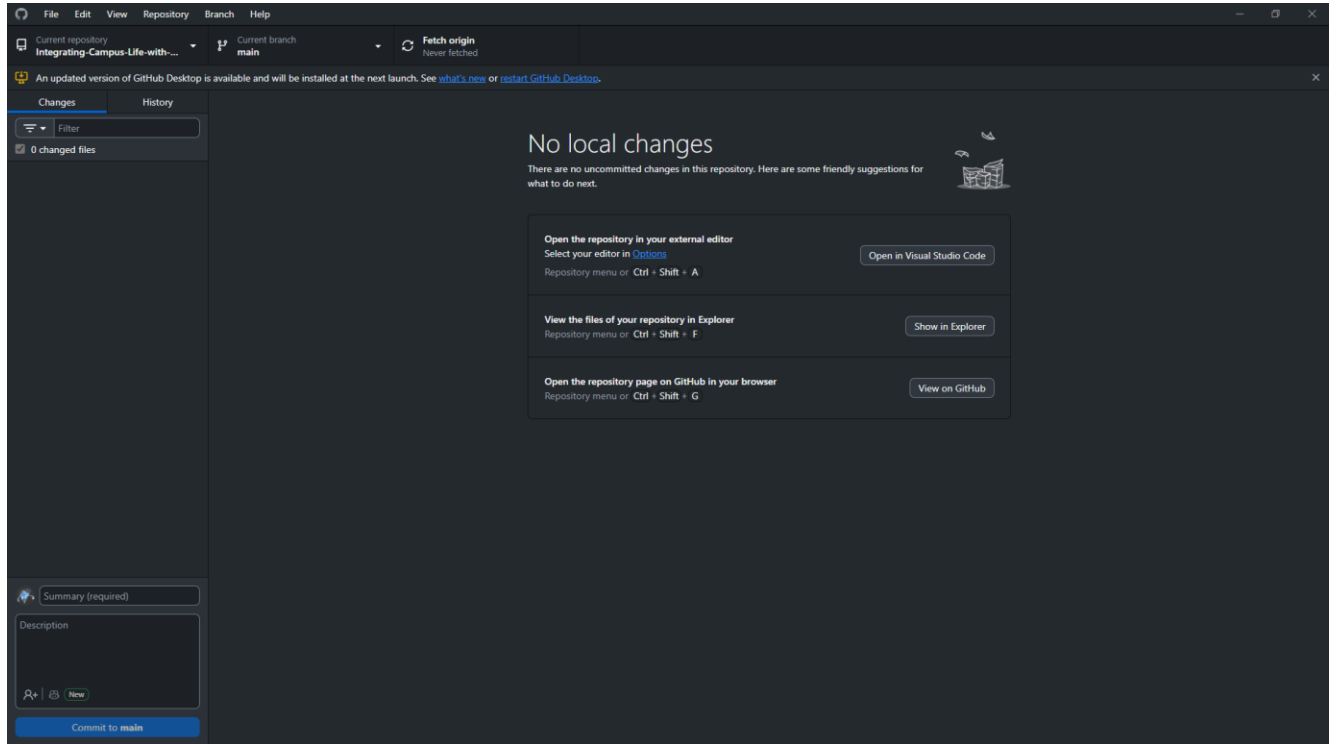


Figure: Printing System

5. GIT WORKFLOW



6. TESTING

Planning:

The test plan was designed by the **QA Team** with input from the **Project Manager** and **Development Leads**.

Goals:

- Ensure every functional and non-functional requirement is tested.
- Detect and resolve defects early.
- Coordinate test execution across modules.
- Deliver a secure, user-friendly, and reliable portal.

People Involved in Testing:

- **1 Project Manager** – Oversees process, ensures alignment with goals.
- **1 QA Lead / Test Engineer** – Prepares test cases, manages testing workflow.
- **2 Developers** – Perform module-specific unit and integration tests.
- **1 UI/UX Designer** – Conducts usability and accessibility testing.

Testing Techniques and Methods:

1. Unit Testing

- Focus on testing individual functions or modules in isolation.
- Each class is tested as a single unit.
- **Example:** Check if the login function correctly validates username and password.

2. Integration Testing:

- Focus on verifying interactions between multiple classes, modules, or subsystems.
- **Checks:** Data flows across interfaces, message passing between objects, and combined workflow correctness.
- **Example:** Test that the Complaint Submission module correctly stores a complaint and the Tracking Module can later retrieve and display its status.

3. White-Box Testing

- Focus on internal logic & code coverage.
- **Checks:** Independent paths, conditions (True/False), loop boundaries, internal data structures.
- **Example:** Ensure the password reset module encrypts new passwords and updates the database correctly.

4. Black-Box Testing

- Focus on input/output behavior.
- **Checks:** Missing functions, interface errors, database connectivity, initialization & termination.
- **Example:** Ensure a student can successfully upload and later download a shared file without corruption.

5. Basis Path Testing

- Use flow graphs to derive independent execution paths.
- Measure **cyclomatic complexity** to define sufficient test cases.

6. Regression Testing

- Re-run key test cases after changes to confirm no unintended errors.
- Example: Ensure password reset doesn't break login.

7. Smoke Testing

- Quick daily build validation.
- Example: Run basic login, file upload, and complaint submission checks after each integration.

8. Higher Order Testing

- **Performance Testing:** Check response time of login, file uploads, and registration under load.
- **Stress Testing:** Simulate peak load (e.g., 10,000 students registering simultaneously).
- **Recovery Testing:** Simulate server crash during payment → check if payment state recovers.
- **Security Testing:** Verify login protection, SQL injection prevention, secure file access.
- **Alpha/Beta Testing:** Internal student testers (Alpha) → limited real users (Beta).

Risks & Mitigation:

- **High Load During Registration** → Conduct stress & load testing.
- **Payment Errors** → Test with mock gateways.
- **File Corruption in Sharing** → Add Checksum verification.
- **Unauthorized Access** → Apply penetration and role-based access testing.

Entry & Exit Criteria:

- **Entry:** Stable build available, development completed, test environment ready.
- **Exit:** All critical test cases passed, no major defects open, performance benchmarks met.

Deliverables:

- Test Plan (this document).
- Detailed Test Cases.
- Test Execution Report.
- Bug Reports.
- Final QA Sign-Off.

Test Case

Test Case 1:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: ARNOB SARKER SUPTA		
Test Case ID: 0001		Test Designed date: 31/08/2025		
Test Priority (Low, Medium, High): High		Test Executed By: NILOY PAUL		
Module Name: LOG IN		Test Execution Date: 31/08/2025		
Test Title: User Log In				
Description: USER NEEDS TO LOGIN TO THE PORTAL BY USERNAME AND PASSWORD				
Precondition: USER MUST HAVE A VALID USERNAME AND PASSWORD				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Enter Username 2. Enter Password 3. Press LOG IN Button	Username, Password	Valid	Valid	Pass
Post Condition: User will Log In and go to the home page.				

Test Case 2:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation	Test Designed By: NILOY PAUL
Test Case ID: 0002	Test Designed date: 31/08/2025
Test Priority (Low, Medium, High): Medium	Test Executed By: ARNOB SARKER SUPTA
Module Name: Home Page	Test Execution Date: 31/08/2025
Test Title: LOGOUT from Home Page	
Description: USER NEEDS TO LOGOUT FROM HOME PAGE	
Precondition: USER MUST BE LOG IN	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press LOGOUT Button		Successful	Successful	Pass
Post Condition: User will go back to LOG IN page				

Test Case 3:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: ARNOB SARKER SUPTA		
Test Case ID: 0003		Test Designed date: 31/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: NILOY PAUL		
Module Name: Home Page		Test Execution Date: 31/08/2025		
Test Title: Profile Information Checking from Home Page				
Description: USER NEEDS TO CHECK PROFILE FROM HOME PAGE				
Precondition:				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press the Profile button		Successful	Successful	Pass
Post Condition: User will go to profile where they will see profile information.				

Test Case 4:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation	Test Designed By: NILOY PAUL
Test Case ID: 0004	Test Designed date: 31/08/2025

Test Priority (Low, Medium, High): Medium		Test Executed By: ARNOB SARKER SUPTA		
Module Name: Change Password		Test Execution Date: 31/08/2025		
Test Title: Changing Password from Home Page				
Description: USER NEEDS TO CHANGE PASSWORD				
Precondition: USER MUST REMEMBER CURRENT PASSWORD				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press Settings button 3. Press Change Password option 4. Enter Current Password 5. Enter NEW PASSWORD 6. Enter CONFIRM NEW PASSWORD 7. Press CHANGE PASSWORD button		Successful	Successful	Pass
Post Condition: USER needs to LOG IN with the new password				

Test Case 5:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation	Test Designed By: ABDULLAH AL TAMJID
Test Case ID: 0005	Test Designed date: 29/08/2025
Test Priority (Low, Medium, High): Medium	Test Executed By: SADIA ISLAM
Module Name: LOST AND FOUND	Test Execution Date: 29/08/2025

Test Title: Testing Report Item in Lost and Found				
Description: USER needs to report an item in lost and found				
Precondition (If any): USER MUST HAVE THE LOCATION AND CLEAR PHOTO OF THE ITEM				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press Report Item button 3. Enter Description 4. Enter Location 5. Provide Photo		Reported	Reported	Pass
Post Condition: Admin approval required for public visibility				

Test Case 6:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: SADIA ISLAM		
Test Case ID: 0006		Test Designed date: 29/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: ABDULLAH AL TAMJID		
Module Name: LOST AND FOUND		Test Execution Date: 29/08/2025		
Test Title: Testing Search Item in Lost and Found				
Description: USER needs to search for items in lost and found				
Precondition (If any): USER MUST KNOW ABOUT THE LOST ITEM				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. LOG IN to the system 2. Enter Item Name 3. Press Search Item button		Successful	Successful	Pass
Post Condition: It will show related items.				

Test Case 7:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: ABDULLAH AL TAMJID		
Test Case ID: 0007		Test Designed date: 29/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: SADIA ISLAM		
Module Name: File a Complaint		Test Execution Date: 29/08/2025		
Test Title: Testing anonymous complaint from submission				
Description: IF USER WANTS, THEY CAN FILE AN ANONYMOUS COMPLAIN FORM				
Precondition (If any): USER MUST KNOW THE COMPLAINT TYPE				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Enter Complaint type 3. Enter Complaint description 4. Upload a file 5. Press Submit button		Submitted	Submitted	Pass
Post Condition: Complaint ID will auto generate				

Test Case 8:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation			Test Designed By: SADIA ISLAM	
Test Case ID: 0008			Test Designed date: 29/08/2025	
Test Priority (Low, Medium, High): Medium			Test Executed By: ABDULLAH AL TAMJID	
Module Name: File a Complain			Test Execution Date: 29/08/2025	
Test Title: Tracking a complaint status				
Description: USER NEEDS TO TRACK A COMPLAINT STATUS				
Precondition (If any): USER MUST HAVE A COMPLAINT ID				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Enter complaint ID 3. Press Track Complaint button		Found	Found	Pass
Post Condition: It will show the status of the complaint as Pending/ In Review/ Resolved.				

Test Case 9:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation	Test Designed By: ABDULLAH AL TAMJID
Test Case ID: 0009	Test Designed date: 29/08/2025
Test Priority (Low, Medium, High): Medium	Test Executed By: SADIA ISLAM
Module Name: File Sharing	Test Execution Date: 29/08/2025
Test Title: Testing drag and drop in File Sharing	

Description: USER NEEDS TO DRAG AND DROP FILE IN FILE SHARING				
Precondition (If any):				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Drag a file and drop it in the section 3. Enter File title 4. Enter Description 5. Select course/group/faculty 6. Enter expiry date 7. Select Permission Settings 8. Press UPLOAD button		Uploaded	Uploaded	Pass
Post Condition: The file will be shown in My Shared Files.				

Test Case 10:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation	Test Designed By: KAHIUM AHAMED FAHIM
Test Case ID: 0010	Test Designed date: 30/08/2025
Test Priority (Low, Medium, High): Medium	Test Executed By: NILOY PAUL
Module Name: File Sharing	Test Execution Date: 30/08/2025
Test Title: Removing a file from my shared files	
Description: USER NEEDS TO REMOVE A FILE FROM MY SHARED FILES SECTION	
Precondition (If any): USER MUST HAVE FILES IN MY SHARED FILES SECTION	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press Remove button		Removed	Removed	Pass
Post Condition: File will be removed from My Shared Files				

Test Case 11:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: NILOY PAUL		
Test Case ID: 0011		Test Designed date: 30/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: KAHNUM AHAMED		
		FAHIM		
Module Name: File Sharing		Test Execution Date: 30/08/2025		
Test Title: Coping a file from my shared files				
Description: USER NEEDS TO COPY A FILE FROM MY SHARED FILES SECTION				
Precondition (If any): USER MUST HAVE FILES IN MY SHARED FILES SECTION				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press Copy button		Copied	Copied	Pass
Post Condition: File will be copied on the clip board.				

Test Case 12:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: KAHIUM AHAMED FAHIM		
Test Case ID: 0012		Test Designed date: 30/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: NILOY PAUL		
Module Name: File Sharing		Test Execution Date: 30/08/2025		
Test Title: Searching for a file from my shared files				
Description: USER NEEDS TO SEARCH FOR A FILE IN MY SHARED FILES SECTION				
Precondition (If any): USER MUST HAVE FILES IN MY SHARED FILES SECTION				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Enter name 3. Select Course 4. Select file type 5. Select DATE SHARED 6. Select Short by		Found	Found	Pass
Post Condition: It will show related files				

Test Case 13:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: NILOY PAUL		
Test Case ID: 0013		Test Designed date: 30/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: KAHIUM AHAMED FAHIM		
Module Name: GYM Registration		Test Execution Date: 30/08/2025		

Test Title: Testing GYM Registration				
Description: USER NEEDS TO REGISTER FOR THE GYM				
Precondition (If any):				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press Register button 3. Complete Payment		Registered	Registered	Pass
Post Condition:				

Test Case 14:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: KAHIMUM AHAMED FAHIM		
Test Case ID: 0014		Test Designed date: 30/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: NILOY PAUL		
Module Name: GYM Registration		Test Execution Date: 30/08/2025		
Test Title: Testing successful payment in GYM registration				
Description: USER NEEDS TO COMPLETE THE PAYMENT FOR GYM REGISTRETION				
Precondition (If any): USER MUST HAVE SUFFICIENT BALANCE IN THEIR ACCOUNT				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. LOG IN to the system 2. Enter Username 3. Select Payment method 4. Enter account number 5. Enter Verification code 6. Enter PIN 7. Press proceed button		Successful	Successful	Pass
Post Condition: After Successful payment gym registration will be completed.				

Test Plan 15:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: NILOY PAUL		
Test Case ID: 0015		Test Designed date: 30/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: KAHIUM AHAMED FAHIM		
Module Name: Faculty Information		Test Execution Date: 30/08/2025		
Test Title: Searching Faculty Information				
Description: USER NEEDS TO SEARCH FACULTY INFORMATION				
Precondition (If any): USER MUST KNOW FACULTY NAME				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Enter Faculty Name 3. Press the search button		Found	Found	Pass
Post Condition: It will show all information about a faculty.				

Test Case 16:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: KAHIMUM AHAMED FAHIM		
Test Case ID: 0016		Test Designed date: 30/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: NILOY PAUL		
Module Name: E-Book & Resources		Test Execution Date: 30/08/2025		
Test Title: Searching Books from Course Book Reference				
Description: USER NEEDS TO SEARCH A PERTICULER COURSE BOOKS				
Precondition (If any): USER MUST KNOW COURSE NAME				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Enter Course Name 3. Press Search button		Found	Found	Pass
Post Condition: It will show all books related to that course.				

Test Case 17:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation	Test Designed By: NILOY PAUL
Test Case ID: 0017	Test Designed date: 30/08/2025
Test Priority (Low, Medium, High): Medium	Test Executed By: KAHIUM AHAMED FAHIM
Module Name: E-Book & Resources	Test Execution Date: 30/08/2025
Test Title: Downloading Books from Course Book Reference	

Description: USER NEEDS TO DOWNLOAD A PERTICULER COURSE BOOKS				
Precondition (If any):				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press Download button		Downloaded	Downloaded	Pass
Post Condition: The books will be downloaded				

Test Case 18:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: ARNOB SARKER SUPTA		
Test Case ID: 0018		Test Designed date: 30/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: ABDULLAH AL TAMJID		
Module Name: AIUB Print		Test Execution Date: 30/08/2025		
Test Title: Testing AIUB printing feature				
Description: USER NEEDS TO PRINT FROM AIUB PRINT				
Precondition (If any): USER MUST HAVE BALANCE IN THEIR ACCOUNT				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Enter the file 3. Select the printer 4. Press Print button		Successful	Successful	Pass
Post Condition: It will deduct some amount of money from the account				

Test Case 19:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: ABDULLAH AL TAMJID		
Test Case ID: 0019		Test Designed date: 30/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: ARNOB SARKER SUPTA		
Module Name: AIUB Print		Test Execution Date: 30/08/2025		
Test Title: Recharging amount in the printing account				
Description: USER NEED TO RECHARG AMOUNT IN THEIR ACCOUNT				
Precondition (If any): USER MUST HAVE SUFFICIENT BALANCE IN THEIR MOBILE BANKING ACCOUNT				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press Recharge button 3. Select Payment method 4. Enter account number 5. Enter Verification code 6. Enter PIN 7. Press proceed button		Successful	Successful	Pass
Post Condition: It will add the amount to the printing account.				

Test Case 20:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation	Test Designed By: ARNOB SARKER SUPTA
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Test Case ID: 0020		Test Designed date:		
Test Priority (Low, Medium, High): Medium		Test Executed By: ABDULLAH AL TAMJID		
Module Name: Notice		Test Execution Date:		
Test Title: Searching for Upcoming Events notice				
Description: USER NEEDS TO SEARCH FOR UPCOMING ENENTS				
Precondition (If any):				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press UPCOMING button		Found	Found	Pass
Post Condition: It will show all u	upcoming events.			

Test Case 21:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation			Test Designed By: ABDULLAH AL TAMJID		
Test Case ID: 0021			Test Designed date: 30/08/2025		
Test Priority (Low, Medium, High): Medium			Test Executed By: ARNOB SARKER SUPTA		
Module Name: Notice			Test Execution Date: 30/08/2025		
Test Title: Searching for Previous Events notice					
Description: USER NEEDS TO SEARCH FOR PREVIOUS ENENTS					
Precondition (If any):					
Test Steps		Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. LOG IN to the system 2. Press PREVIOUS button		Found	Found	Pass
Post Condition: It will show all previous events.				

Test Case 22:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: ABDULLAH AL TAMJID		
Test Case ID: 0022		Test Designed date: 30/08/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: ARNOB SARKER SUPTA		
Module Name: Notice		Test Execution Date: 30/08/2025		
Test Title: Searching Notice by date				
Description: USER NEEDS TO SEARCH NOTICE BY DATE				
Precondition (If any): USER MUST KNOW THE DATE				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Enter the date 3. Press the search button		Found	Found	Pass
Post Condition: It will show all related Notice for the day.				

Test Case 23:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation			Test Designed By: ARNOB SARKER SUPTA	
Test Case ID: 0023			Test Designed date: 30/08/2025	
Test Priority (Low, Medium, High): Medium			Test Executed By: ABDULLAH AL TAMJID	
Module Name: Registration			Test Execution Date: 30/08/2025	
Test Title: Testing Registration				
Description: USER NEEDS TO COMPLETE THEIR REGISTRATION				
Precondition (If any): USER MUST COMPLETE PRE-REQUISITE				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press Academics button 3. Press Registration button		Successful	Successful	Pass
Post Condition: It will show the registration procedure.				

Test Case 24:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation	Test Designed By: NILOY PAUL
Test Case ID: 0024	Test Designed date: 01/09/2025
Test Priority (Low, Medium, High): Medium	Test Executed By: ARNOB SARKER SUPTA
Module Name: Location Tracking	Test Execution Date: 01/09/2025
Test Title: Turning on current location	

Description: USER NEEDS TO TURN ON CURRENT LOCATION FOR LOCATION ACCESS				
Precondition (If any): USER MUST TURN ON THEIR GPS				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press Location button		Successful	Successful	Pass
Post Condition: It will turn on current location.				

Test Case 25:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: ARNOB SARKER SUPTA		
Test Case ID: 0025		Test Designed date: 01/09/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: NILOY PAUL		
Module Name: Location Tracking		Test Execution Date: 01/09/2025		
Test Title: Testing Destination Located				
Description: USER NEEDS TO INPUT DESTINATION				
Precondition (If any): USER MUST ENABLE THEIR LOCATION				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Enter Destination		Successful	Successful	Pass
Post Condition: It will locate the destination.				

Test Case 26:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation			Test Designed By: NILOY PAUL		
Test Case ID: 0026			Test Designed date: 01/09/2025		
Test Priority (Low, Medium, High): Medium			Test Executed By: ARNOB SARKER SUPTA		
Module Name: Location Tracking			Test Execution Date: 01/09/2025		
Test Title: Testing navigation Tracking					
Description: USER NEEDS TO GET THE NAVIGATION FOR THE DESTINATION					
Precondition (If any): USER MUST ENABLE THEIR LOCATION AND ENTERED THEIR DESTINATION					
Test Steps		Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press the Located me button			Successful	Successful	Pass
Post Condition: Navigation from current location to the destination will appear.					

Test Case 27:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation	Test Designed By: ARNOB SARKER SUPTA
Test Case ID: 0027	Test Designed date: 01/09/2025
Test Priority (Low, Medium, High): Medium	Test Executed By: NILOY PAUL
Module Name: LOGIN Page	Test Execution Date: 01/09/2025
Test Title: Testing can't access your account button	
Description: USER NEEDS TO BE ABLE TO RESET THEIR PASSWORD	

Precondition (If any): USER MUST HAVE A VALIDE USERNAME				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Press can't access your account button 3. Enter Username 4. Enter Captcha 5. Press Next button		Successful	Successful	Pass
Post Condition: It will reset the password.				

Test Case 28:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: NILOY PAUL		
Test Case ID: 0028		Test Designed date: 01/09/2025		
Test Priority (Low, Medium, High): HIGH		Test Executed By: ARNOB SARKER SUPTA		
Module Name: LOG IN		Test Execution Date: 01/09/2025		
Test Title: User Log In				
Description: USER NEEDS TO LOGIN TO THE PORTAL BY USERNAME AND PASSWORD				
Precondition (If any): USER MUST HAVE A VALID USERNAME AND PASSWORD				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Enter Username 2. Enter Password 3. Press LOG IN Button	Username, Password	Valid	Invalid	Fail
Post Condition: User will try again with valid username and password.				

Test Case 29:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: ARNOB SARKER SUPTA		
Test Case ID: 0029		Test Designed date: 01/09/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: NILOY PAUL		
Module Name: GYM Registration		Test Execution Date: 01/09/2025		
Test Title: Testing successful payment in GYM registration				
Description: USER NEEDS TO COMPLETE THE PAYMENT FOR GYM REGISTRETION				
Precondition (If any): USER MUST HAVE SUFFICIENT BALANCE IN THEIR ACCOUNT				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Enter Username 3. Select Payment method 4. Enter account number 5. Enter Verification code 6. Enter PIN 7. Press proceed button		Successful	Unsuccessful	Fail
Post Condition: User’s payment request will be canceled. User needs to request again.				

Test Case 30:

Project Name: AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation		Test Designed By: NILOY PAUL		
Test Case ID: 0030		Test Designed date: 01/09/2025		
Test Priority (Low, Medium, High): Medium		Test Executed By: ARNOB SARKER SUPTA		
Module Name: AIUB Print		Test Execution Date: 01/09/2025		
Test Title: Testing AIUB printing feature				
Description: USER NEEDS TO PRINT FROM AIUB PRINT				
Precondition (If any): USER MUST HAVE BALANCE IN THEIR ACCOUNT				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. LOG IN to the system 2. Enter the file 3. Select the printer 4. Press Print button		Successful	Unsuccessful	Fail
Post Condition: The printing request will be canceled. User needs to select properly and try again.				

7. SOFTWARE PRODUCT METRICS

7.1 Function-based Metrics (Function Points Analysis)

Function Points (FP) measure functionality delivered to the user by identifying **Number of External Inputs (EI)**, **Number of External Output (EO)**, **Number of external inquiries (EQ)**, **Number of internal files (ILF)**, **Number of external interfaces (EIF)**.

- **Number of External Inputs (EI) (user data entry):** Login, course registration, payment form, grievance submission → ~10
- **Number of External Output (EO) (reports, displays):** Grade reports, event notices, fee receipts, library records → ~8
- **Number of internal files (ILF) (logical data groups):** Student info, course catalog, faculty data, alumni, payments, library → ~6
- **Number of external interfaces (EIF) (external systems):** Payment gateway, email system, university DB → ~3
- **Number of external inquiries (EQ) (user queries):** Search courses, view schedule, check fees → ~5

Function Point (FP) calculation

- **External Inputs (EI):** $32 \times 4 = 128$
- **External Outputs (EO):** $60 \times 5 = 300$
- **External Inquiries (EQ):** $24 \times 4 = 96$
- **Internal Logical Files (ILF):** $8 \times 10 = 80$
- **External Interface Files (EIF):** $2 \times 7 = 14$

Count Total = $128 + 300 + 96 + 80 + 14 = 618$

Complexity Adjustment Factor (CAF)

$CAF = 0.65 + 0.01 \times 70 = 1.35$

Function Points (FP)

$FP = \text{Count Total} \times CAF = 618 \times 1.35 = 834.3$

Final FP = ~834 Function Points

This project delivers the equivalent functionality of about 834 FP, which is quite a large system in terms of size and scope.

7.2 Object-Oriented & Class Metrics

Using OO metrics :

- **WMC (Weighted Methods per Class):**
 - Example: User class = 8 methods (login, register, update, delete, etc.)
 - Average across project $\sim 10 \rightarrow$ manageable complexity.
- **DIT (Depth of Inheritance Tree):**
 - Example: Admin $\rightarrow$ Faculty $\rightarrow$ Student hierarchies.
 - Average depth = 2 $\rightarrow$ good reuse, not too deep.
- **NOC (Number of Children):**
 - Example: User superclass has Student, Faculty, Admin (NOC = 3).
 - Indicates moderate use of inheritance.
- **CBO (Coupling Between Objects):**
 - High coupling between Registration $\leftrightarrow$ Payment modules.
 - Average CBO $\sim 4\text{--}5 \rightarrow$ acceptable, but Payment should be isolated.
- **LCOM (Lack of Cohesion of Methods):**
 - Cohesion high in Student/Faculty modules (methods relate well).
 - LCOM ~ 0.3 (low $\rightarrow$ good cohesion).

System shows **balanced OOP design** with manageable complexity, moderate coupling, and high cohesion.

7.3 Operation-oriented Metrics

- **Average operation size:**
 - Most operations handle $\sim 30\text{--}50$ lines of code $\rightarrow$ moderate.
- **Average complexity (Cyclomatic Complexity):**
 - Login method ~ 2 , Course Registration ~ 5 , Payment processing ~ 7 (higher due to conditions).
 - Average CC = $4\text{--}5 \rightarrow$ system is maintainable.

- **Number of parameters per operation:**
 - Login (2), Course Registration (3–4), Payment (5).
 - Average = 3 → reasonable, not overloaded.

Operations are moderately complex and efficient, avoiding too many parameters per function.

4. Maintenance Metrics (Software Maturity Index – SMI)

Formula:

$$SMI = \frac{MT - (f_a + f_c + f_d)}{MT}$$

Where:

- M_T = Total modules in current release.
- f_a = Modules added.
- f_c = Modules changed.
- f_d = Modules deleted.

Example (for your portal after first release):

$M_T = 50$ modules.

$f_a = 9$, $f_c = 2$, $f_d = 0$

$$SMI = \frac{MT - (f_a + f_c + f_d)}{MT} = \frac{50 - (9 + 2 + 0)}{50} = 0.78$$

$SMI = 0.78 \rightarrow$ System is **reasonably stable**, but moderate changes exist due to evolving requirements.

8. CONCLUSION AND FUTURE WORK

8.1.1 Conclusion

The AIUB Portal Redefined project successfully integrates academic, administrative, and student-support services into a single, smart, and user-friendly digital platform. By adopting the SCRUM process model, the project-maintained adaptability, continuous feedback, and incremental development, ensuring that complex modules such as CGPA-based registration, AR navigation, cloud printing, file sharing, and anonymous complaint systems were designed effectively.

Through rigorous software engineering practices—requirements analysis, design modelling, testing, risk management, and metrics evaluation—the project demonstrated scalability, usability, and reliability. The testing phase validated that the system fulfils both functional and non-functional requirements, while project estimation and scheduling ensured realistic resource allocation and timeline management.

Overall, this project contributes to AIUB’s vision of a technologically advanced, student-centric environment by reducing inefficiencies, improving accessibility, and enhancing engagement across the campus community. It provides not only a technical solution but also addresses societal, cultural, and legal considerations such as inclusivity, data privacy, and equitable access.

8.1.2 Future Work

While the project establishes a solid foundation, several areas can be explored to further enhance its impact:

1. **AI/ML Expansion** – Enhance course demand prediction models with advanced machine learning algorithms for better accuracy and personalization.
2. **Mobile App Development** – Develop a resolute Android and iOS app for improved accessibility and real-time notifications.
3. **Integration with External Systems** – Connect the portal with external platforms such as digital libraries, online learning systems, and government education databases.
4. **Blockchain for Security** – Implement blockchain-based solutions for secure record-keeping in areas like registration logs and complaint handling.
5. **IoT Integration** – Expand AR navigation with IoT devices like smart sensors for real-time crowd management and emergency alerts.
6. **Scalability Testing at Large Scale** – Conduct stress testing with a full-scale user base (e.g., all AIUB students simultaneously) to ensure 24/7 reliability.
7. **Continuous Feedback System** – Introduce in-app surveys and AI-based chatbots to capture user feedback and resolve issues promptly.
8. **Sustainability Features** – Incorporate eco-friendly features, such as tracking paper usage from printing and encouraging digital alternatives.

By addressing these areas, the portal can evolve into a comprehensive digital ecosystem that not only supports AIUB’s academic operations but also redefines campus life through innovation, inclusivity, and long-term sustainability.